

MIAE 221: Materials Science for Engineers

Callister Textbook Companion & Ultimate Lecture Study Guide

Concordia University · Gina Cody School of Engineering and Computer Science

Textbook:

Materials Science and Engineering: An Introduction

(10th Edition, Callister & Rethwisch)

Instructor Alignment: Dr. Mamoun Medraj, P.Eng | **Scope:** Complete Course Syllabus (Chapters 1–7, 9, 12–15, 18, 19, 21)



Executive Overview & How to Use This Guide

This Master Study Guide bridges the **official Callister & Rethwisch textbook** with **Dr. Mamoun Medraj's lecture slides, tutorial problem sets, and Concordia examination expectations**.

Instead of wading through nearly 1,000 pages of dense textbook text, this guide condenses every syllabus topic into a structured **5-Part Mastery Architecture**:

1. 🎯 **30-Second Core Intuition & Plain-English Concept:** The fundamental physical reality explained without jargon.
 2. ⚙️ **High-Yield Mathematical Engine & Essential Laws:** Formulas, exact variable definitions, and SI units.
 3. 🖼️ **Textbook Reference Diagram & Visual Pedagogical Analysis:** High-resolution diagrams extracted directly from Callister & Rethwisch (or Dr. Medraj's slides) with step-by-step visual breakdowns.
 4. ⚖️ **Teacher Notes Cross-Reference & Concordia Exam Traps:** Detailed comparison between Callister's textbook theory and Dr. Medraj's specific lecture slides, tutorial quizzes, and midterm/final exam traps.
 5. 💡 **Master Exam Problem & Step-by-Step Solution Framework:** A representative quantitative or conceptual problem solved with the exact step-by-step methodology required on Concordia exams.
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Master Alignment Matrix: Callister Chapters vs. Dr. Medraj Lectures

Callister Chapter	Topic Title	Dr. Medraj Lecture	Course Week	Key Formula / Core Concept	Concordia Exam Focus
Ch 1	Introduction & Materials Classification	Lecture 1	Week 1	Materials Tetrahedron ($P \rightarrow S \rightarrow P \rightarrow P$)	Material classes, Liberty ships case study
Ch 2	Atomic Structure & Bonding	Lectures 2 & 3	Weeks 1–2	$E_N = -\frac{A}{r} + \frac{B}{r^n}$	Force balance, energy wells, ionic character
Ch 3	Structure of Crystalline Solids	Lectures 4, 5 & 6	Weeks 2–3	$\rho = \frac{nA}{V_c N_A}$, $n\lambda = 2d_{hkl} \sin \theta$	APF, $[uvw]$, (hkl) , Linear/Planar density, XRD
Ch 4	Imperfections in Solids	Lecture 7	Week 4	$N_v = Ne^{-Q_v/k_B T}$, Hume-Rothery rules	Vacancy Arrhenius plots, Burgers vector $\vec{b}$
Ch 5	Diffusion in Solids	Syllabus Ch 5	Week 5	$J = -D \frac{dC}{dx}$, $\frac{C_s - C_0}{C_s - C_0} = 1 - \operatorname{erf}\left(\frac{x}{2\sqrt{Dt}}\right)$	Carburization depth, $x^2/(Dt) = \text{const}$, Arrhenius D
Ch 6	Mechanical Properties of Metals	Syllabus Ch 6	Week 6	$\sigma = E\epsilon$, 0.2 Yield σ_y , $\sigma_T = \sigma(1 + \epsilon)$	Engineering vs True $\sigma - \epsilon$, Resilience, Toughness
Ch 7	Dislocations & Strengthening	Syllabus Ch 7.1–7.4	Week 7	$\tau_R = \sigma \cos \phi \cos \lambda$, $\sigma_y = \sigma_0 + k_y d^{-1/2}$	Schmid's law, Hall-Petch, Annealing stages
Ch 9	Phase Diagrams & $Fe - Fe_3C$	Syllabus Ch 9	Week 8	Lever Rule: $W_L = \frac{C_\alpha - C_0}{C_\alpha - C_L}$, $P + F = C + 1$	$Fe - Fe_3C$ eutectoid, Pearlite, Proeutectic phases
Ch 12 & 13	Ceramics: Structures & Properties	Syllabus Ch 12.3–12.11	Week 9	r_C/r_A coordination, $\sigma_{fs} = \frac{3F_f L}{2bd^2}$	Rock Salt, Perovskite, Silicate glass, 3-point bend
Ch 14 & 15	Polymers: Structures & Properties	Syllabus Ch 14 & 15.1–14	Week 10	$DP = M_n/m$, $PDI = M_w/M_n$, T_g vs T_m	Thermoplastics vs Thermosets, Viscoelasticity
Ch 18	Electrical Properties	Syllabus Ch 18	Week 11	$\sigma = ne\mu$	e
Ch 19	Thermal Properties	Syllabus Ch 19	Week 11	$\frac{\Delta l}{l_0} = \alpha_l \Delta T$, $TSR \cong \frac{\sigma_f k}{E \alpha_l}$	Potential well asymmetry, Thermal shock resistance
Ch 21	Optical Properties	Syllabus Ch 21	Week 12	$h\nu = hc/\lambda \geq E_g$, $R + A + T = 1$	Photon absorption vs transparency in insulators

Chapter 1: Introduction to Materials Science and Engineering

(Aligned with Dr. Medraj Lecture 1 · Week 1)

1. 🎯 30-Second Core Intuition & Plain-English Concept

Materials Science investigates **why** materials behave the way they do by examining atomic and crystal structures. Materials Engineering focuses on **how** to manipulate those structures to design useful products.

The entire discipline rests upon the **Materials Tetrahedron**:

Processing → Structure → Properties → Performance

Real-World Analogy

: Making pizza. The ingredients and baking temperature (**Processing**) dictate whether the crust is doughy or crispy (**Structure**). That internal texture determines how stiff the slice is under gravity (**Properties**), which decides whether your toppings slide into your lap or hold firm (**Performance**).

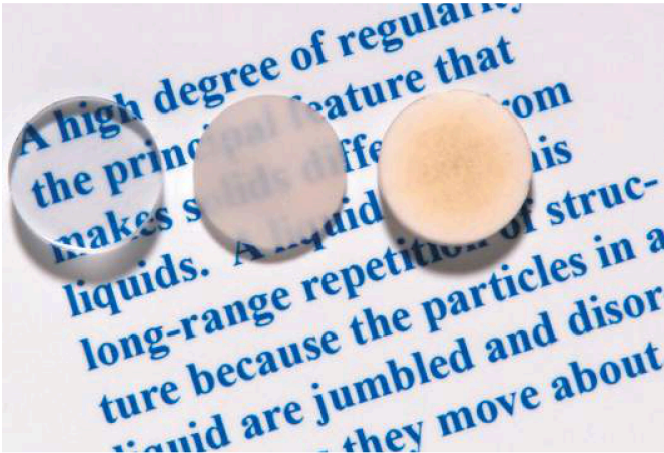
2. **High-Yield Key Information & Classification Matrix**

Engineering materials are divided into five fundamental classes based on atomic bonding and macroscopic behavior:

Material Class	Dominant Bonding	Key Distinctive Properties	Typical Engineering Applications
Metals	Metallic (electron sea)	High electrical/thermal conductivity, ductile, high stiffness, reflective	Structural airframes, engine crankshafts, electrical wiring
Ceramics	Ionic & Covalent	Extreme hardness, high melting temperature, electrically insulating, brittle	Thermal barrier coatings on turbine blades, spark plugs, abrasive cutting tools
Polymers	Covalent backbone + Secondary van der Waals	Low density, flexible, low melting point, corrosion resistant	Automotive bumpers, beverage bottles, biomedical catheters
Composites	Multi-phase mixture	Tailored property combinations (high strength-to-weight ratio)	Carbon fiber reinforced epoxy wings (Boeing 787), fiberglass boats
Advanced Materials	Mixed (Semiconductors, Biomaterials, Smart)	Highly tailored electrical, biocompatible, or sensory response	Microprocessors (Si), shape memory alloys (Nitinol stents), piezoelectric sensors

3. **Textbook Reference Diagram & Visual Pedagogical Analysis**

Figure 1.1 Three thin disk specimens of aluminum oxide that have been placed over a printed page in order to demonstrate their differences in light-transmittance characteristics. The disk on the left is *transparent* (i.e., virtually all light that is reflected from the page passes through it), whereas the one in the center is *translucent* (meaning that some of this reflected light is transmitted through the disk). The disk on the right is *opaque*—that is, none of the light passes through it. These differences in optical properties are a consequence of differences in structure of these materials, which have resulted from the way the materials were processed.



William D. Callister Jr./ Specimen preparation,

Figure 1.1: Three thin disk specimens of aluminum oxide (Al_2O_3) demonstrating the profound effect of internal structure on optical properties.



Figure 1.2 The four components of the discipline of materials science and engineering and their interrelationship.

Figure 1.2: The four interconnected pillars of materials science and engineering.

In-Depth Visual Breakdown:

- **The Alumina Disks (Figure 1.1):** All three disks have the exact identical chemical formula (Al_2O_3), yet their optical properties are vastly different:
 1. *Left disk (Transparent)*
: Single crystal sapphire. No internal grain boundaries or voids; photons pass directly through without scattering.
 2. *Center disk (Translucent)*
: Dense polycrystalline alumina. Light scatters at the boundaries between randomly oriented microcrystals.
 3. *Right disk (Opaque)*
: Sintered polycrystalline alumina containing residual microscopic pores. Pores have a drastically different refractive index than alumina, scattering light completely and making the disk appear milky white.
- **Core Takeaway:** Changing the **processing** method altered the internal **structure** (single crystal vs. polycrystalline vs. porous), which directly altered the optical **property** (transparency), dictating **performance** (optical window vs. opaque insulator).

4. 🏛️ Teacher Notes Cross-Reference & Concordia Exam Traps

- **What Callister Emphasizes:** General historical evolution (Stone Age → Bronze Age → Iron Age) and broad societal sustainability.
- **What Dr. Medraj Emphasizes in Lecture 1:**
 - **The Liberty Ship Failures Case Study:** During WWII, all-welded Liberty ships fractured catastrophically in cold North Atlantic waters. Dr. Medraj specifically highlights three structural causes:
 1. *Temperature*
: Steel went below its **Ductile-to-Brittle Transition Temperature (DBTT)**.
 2. *Stress Concentrations*
: Square hatch corners acted as severe stress risers where cracks initiated.
 3. *Processing (Welding vs. Riveting)*
: Riveted plates stop running cracks at the plate joint; continuous welded hulls allowed a brittle crack to propagate uninterrupted through the entire circumference of the ship.
 - **Beverage Container Material Selection:** Why is a soda can aluminum (ductile, recyclable, impervious to gas, cold-drawn), while a wine bottle is glass (chemically inert, rigid, non-permeable), and a water bottle is PET (lightweight, shatterproof)?
- **Concordia Exam Trap:** Confusing **stiffness** (elastic modulus E) with **strength** (yield stress σ_y). A ceramic has a higher stiffness than most metals, but lower tensile strength because of flaw-induced brittle fracture!

5. 📌 Master Concept Problem & Solution Framework

Exam Question:

Explain, using the materials science tetrahedron, why cold-rolled aluminum sheet exhibits higher yield strength than furnace-annealed aluminum sheet of identical chemical composition.

- **Step 1 (Identify Processing Difference):** Cold rolling plastically deforms the aluminum at room temperature, while annealing involves heating the metal above its recrystallization temperature.
- **Step 2 (Map to Internal Structure):** Cold rolling multiplies dislocation density by several orders of magnitude ($10^5 \rightarrow 10^{10} \text{ cm}^{-2}$) and creates tangled dislocation networks. Annealing provides thermal energy for recovery and recrystallization, producing defect-free equiaxed grains with low dislocation density.
- **Step 3 (Map to Property):** Moving dislocations become tangled and pin one another in the cold-rolled metal (strain hardening), requiring much higher shear stress to initiate further plastic flow. Yield strength σ_y increases significantly.
- **Step 4 (Conclusion):** Different processing $\rightarrow$ different dislocation structure $\rightarrow$ higher yield strength.

📖 Chapter 2: Atomic Structure and Interatomic Bonding

(Aligned with Dr. Medraj Lectures 2 & 3 · Weeks 1–2)

1. 🎯 30-Second Core Intuition & Plain-English Concept

Why do solid materials not collapse into nothingness or fly apart into space? Because atoms operate under a constant energetic balance between **electrostatic attraction** (pulling oppositely charged electrons and nuclei together) and **Pauli repulsion** (electron clouds resisting overlap).

Solid materials settle at the exact separation distance (r_0) where the net interatomic force is **zero** and the net potential energy is at a **minimum**.

Real-World Analogy

: Two billiard balls connected by a stiff spring. If you pull them apart, tension pulls them back (attractive force). If you shove them into each other, the spring fiercely pushes back (repulsive force). At rest, they sit at the equilibrium spring length (r_0).

2. ⚙️ High-Yield Mathematical Engine & Essential Laws

1. Quantum Numbers & Shell Capacities

- n (Principal): Shell energy level ($n = 1, 2, 3, 4, \dots$).
- l (Azimuthal / Subshell): Orbit shape ($l = 0 (s), 1 (p), 2 (d), 3 (f); 0 \leq l \leq n - 1$).
- m_l (Magnetic): Spatial orientation ($-l \leq m_l \leq +l; 2l + 1$ orbitals per subshell).
- m_s (Spin): Electron spin ($+1/2, -1/2$).

Total electrons per shell

: $2n^2$.

2. Net Potential Energy and Interatomic Force

$$E_N(r) = E_A(r) + E_R(r) = -\frac{A}{r^m} + \frac{B}{r^n} \quad (m = 1 \text{ for simple ions, } n \approx 8 - 12)$$

$$F_N(r) = -\frac{dE_N}{dr} = -\frac{mA}{r^{m+1}} + \frac{nB}{r^{n+1}}$$

3. Equilibrium Conditions at $r = r_0$

$$F_N(r_0) = 0 \iff F_A(r_0) = -F_R(r_0)$$

$$\left. \frac{dE_N}{dr} \right|_{r=r_0} = 0 \implies r_0 = \left(\frac{nB}{mA} \right)^{\frac{1}{n-m}}$$

$$E_0 = E_N(r_0) = -\frac{A}{r_0} \left(1 - \frac{1}{n} \right) \quad (\text{for } m = 1)$$

4. Pauling's Percent Ionic Character

Where X_A, X_B are the Pauling electronegativities of the two elements.

3. Textbook Reference Diagram & Visual Pedagogical Analysis

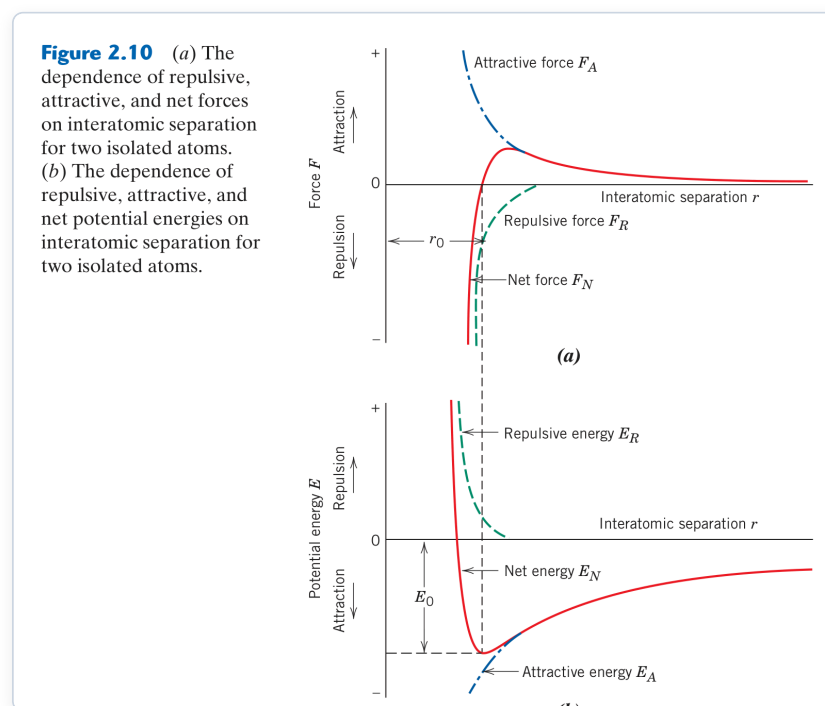


Figure 2.10: (a) Net interatomic force F_N vs. separation distance r . (b) Net potential energy E_N vs. separation distance r , displaying the equilibrium bonding well.

In-Depth Visual Breakdown:

1. Force Curve (Top Plot):

- As $r \rightarrow \infty$, forces approach zero.
- At large r , attractive force F_A dominates (negative slope region).
- At $r = r_0$, the net force curve crosses the horizontal axis: $F_N(r_0) = 0$.
- For $r < r_0$, the curve shoots rapidly into positive territory (repulsive force dominates sharply due to overlapping closed electron shells).

2. Potential Energy Well (Bottom Plot):

- The trough of the curve defines the **Bonding Energy (E_0)** and **Equilibrium Spacing (r_0)**.
- **Physical Property Links:**
 - **Melting Point (T_m):** Proportional to the depth of the well ($|E_0|$). Deeper well = higher thermal energy required to break bonds = high T_m (e.g., Diamond, Tungsten).

- **Elastic Modulus (E):** Proportional to the curvature at the bottom of the well (second derivative $\left. \frac{d^2 E}{dr^2} \right|_{r_0}$).
Steeper curvature = stiffer spring = high Young's modulus.
- **Thermal Expansion Coefficient (α_l):** Governed by the **asymmetry (anharmonicity)** of the well. As temperature rises, atoms oscillate; because the repulsive side is steeper than the attractive side, the mean interatomic distance shifts outward ($r_T > r_0$). Highly symmetric wells exhibit near-zero thermal expansion.

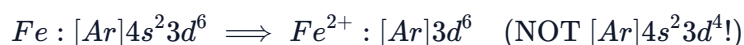
4. 📚 Teacher Notes Cross-Reference & Concordia Exam Traps

- **Dr. Medraj Focus (Lectures 2 & 3):**

- Emphasizes the mathematical derivation connecting Coulomb's law ($F_A = \frac{|z_1 z_2| e^2}{4\pi\epsilon_0 r^2}$) with repulsive power laws.
- Practice Problem Set #1 contains exact calculation questions for $K^+ - Cl^-$ ion pairs and gold wire atomic counts.

- **Concordia Exam Traps:**

- **Transition Metal Ionization Trap:** When writing electron configurations for cations (e.g., Fe^{2+}), electrons are always removed from the outermost valence shell **first**:



- **Force vs. Energy Derivative Sign:** Remember that $F = -\frac{dE}{dr}$. In many physics texts, $F = +\frac{dE}{dr}$ depending on sign conventions. In Callister and Dr. Medraj's slides, attractive force is negative, repulsive is positive, and equilibrium occurs at the zero-crossing.

5. 💡 Master Exam Problem & Step-by-Step Solution Framework

Problem (Practice Problem Set #1 Archetype):

Two isolated ions K^+ and Cl^- experience an attractive potential energy $E_A = -\frac{1.436}{r}$ eV and a repulsive potential energy $E_R = \frac{7.32 \times 10^{-6}}{r^8}$ eV (where r is in nm). Calculate: (a) Equilibrium interatomic separation r_0 , and (b) Bonding energy E_0 .

- **Step 1: Formulate Net Potential Energy**

$$E_N(r) = -1.436r^{-1} + 7.32 \times 10^{-6}r^{-8}$$

- **Step 2: Differentiate with Respect to r and Set to Zero**

$$\frac{dE_N}{dr} = \frac{1.436}{r^2} - 8 \times \frac{7.32 \times 10^{-6}}{r^9} = 0$$

- **Step 3: Solve for Equilibrium Separation r_0**

$$\frac{1.436}{r_0^2} = \frac{5.856 \times 10^{-5}}{r_0^9} \implies r_0^7 = \frac{5.856 \times 10^{-5}}{1.436} = 4.078 \times 10^{-5} \text{ nm}^7$$

$$r_0 = (4.078 \times 10^{-5})^{1/7} = 0.236 \text{ nm}$$

- **Step 4: Compute Bonding Energy $E_0 = E_N(r_0)$**

$$E_0 = -\frac{1.436}{0.236} + \frac{7.32 \times 10^{-6}}{(0.236)^8} = -6.085 \text{ eV} + 0.761 \text{ eV} = -5.324 \text{ eV}$$

$$\text{Bonding energy magnitude } |E_0| = 5.32 \text{ eV}$$

Chapter 3: The Structure of Crystalline Solids

(Aligned with Dr. Medraj Lectures 4, 5 & 6 · Weeks 2–3)

1. 30-Second Core Intuition & Plain-English Concept

Atoms in solids don't arrange randomly (unless they are glass or polymers). Metals arrange in tightly ordered 3D repeating grids called **crystal lattices**. How tightly and efficiently atoms pack determines how dense the metal is, how easily it deforms under a hammer, and how it diffracts X-rays.

Real-World Analogy

: Stacking oranges in a grocery store display. If you stack them directly on top of each other in a grid, you get Simple Cubic (very loose, wobbles easily). If you drop each orange into the valley between oranges in the layer below, you get close-packing (FCC or HCP, rock solid, maximum packing density).

2. High-Yield Mathematical Engine & Metallic Unit Cells

Crystal Structure	Coordination Number (CN)	Atoms per Cell (n)	Lattice Parameter $a(R)$	Atomic Packing Factor (APF)	Close-Packed Direction
Simple Cubic (SC)	6	1	$a = 2R$	$\frac{\pi}{6} \approx 0.524$	$\langle 100 \rangle$
Body-Centered Cubic (BCC)	8	2	$a = \frac{4R}{\sqrt{3}}$	$\frac{\pi\sqrt{3}}{8} \approx 0.680$	$\langle 111 \rangle$
Face-Centered Cubic (FCC)	12	4	$a = 2\sqrt{2}R$	$\frac{\pi\sqrt{2}}{6} \approx 0.740$	$\langle 110 \rangle$
Hexagonal Close-Packed (HCP)	12	6	$a = 2R, c = 1.633a$	$\frac{\pi\sqrt{3}}{8} \approx 0.740$	$\langle 11\bar{2}0 \rangle$

Essential Crystallographic Equations:

1. Theoretical Density:

$$\rho = \frac{n \cdot A}{V_c \cdot N_A}$$

(Units: A in g/mol, V_c in cm^3 , $N_A = 6.022 \times 10^{23}$ atoms/mol, yielding ρ in g/cm^3)

2. Linear Density (LD):

$$\text{LD} = \frac{\text{number of atom diameters centered on direction vector}}{\text{length of direction vector}}$$

3. Planar Density (PD):

$$\text{PD} = \frac{\text{number of atom cross-sectional areas centered on plane}}{\text{area of plane}}$$

4. Bragg's Law for X-Ray Diffraction (XRD):

$$n\lambda = 2d_{hkl} \sin \theta$$

Interplanar spacing for cubic systems

$$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

5. Diffraction Reflection Selection Rules:

- **BCC**: Reflections occur **only** when $h + k + l = \text{even}$ (e.g., (110), (200), (211), (220)).
- **FCC**: Reflections occur **only** when h, k, l are **unmixed** (all odd or all even) (e.g., (111), (200), (220), (311)).

3. 🖼️ Textbook Reference Diagram & Visual Pedagogical Analysis

the atomic hard-sphere model) divided by the unit cell volume—that is,

$$\text{APF} = \frac{\text{volume of atoms in a unit cell}}{\text{total unit cell volume}} \quad (3.3)$$

For the FCC structure, the atomic packing factor is 0.74, which is the maximum packing possible for spheres all having the same diameter. Computation of this APF is also included as an example problem. Metals typically have relatively large atomic packing factors to maximize the shielding provided by the free electron cloud.

The Body-Centered Cubic Crystal Structure

Another common metallic crystal structure also has a cubic unit cell with atoms located at all eight corners and a single atom at the cube center. This is called a **body-centered cubic (BCC)** crystal structure. A collection of spheres depicting this crystal structure is shown in Figure 3.2c, whereas Figures 3.2a and 3.2b are diagrams of BCC unit cells with

Figure 3.2: Body-Centered Cubic (BCC) unit cell: (a) Hard-sphere model, (b) Reduced-sphere unit cell, and (c) Aggregate of atoms highlighting center-corner contact along the body diagonal.

Figure 3.22 Schematic diagram of an x-ray diffractometer; T = x-ray source, S = specimen, C = detector, and O = the axis around which the specimen and detector rotate.

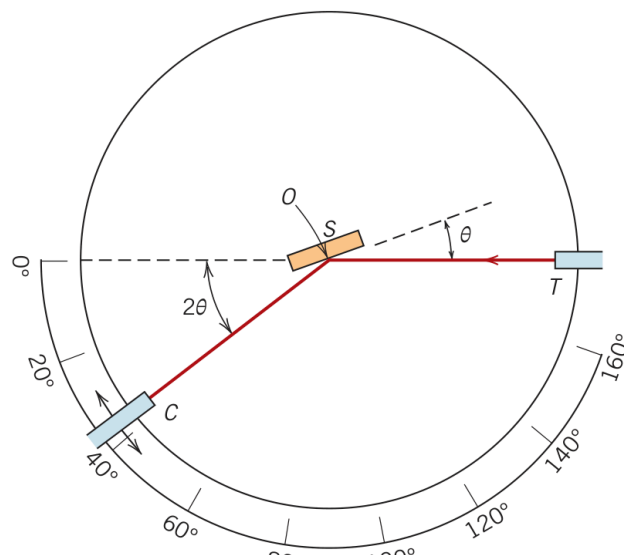


Figure 3.22: Operating geometry of an X-ray diffractometer showing source T , specimen S , and detector C rotating through angle 2θ .

In-Depth Visual Breakdown:

- **BCC Unit Cell (Figure 3.2)**: Notice the hard-sphere contact. Atoms touch strictly along the cube **body diagonal** ([111] direction). Therefore:

$$\text{Body Diagonal} = \sqrt{a^2 + a^2 + a^2} = a\sqrt{3} = 4R \implies a = \frac{4R}{\sqrt{3}}$$

- **X-Ray Diffractometer (Figure 3.22):** Monochromatic X-rays of known wavelength λ strike the powder specimen at angle θ . When the path difference between adjacent crystallographic planes equals an integer number of wavelengths ($n\lambda$), constructive interference produces an intense diffraction peak recorded at detector angle 2θ .

4. 🏛️ Teacher Notes Cross-Reference & Concordia Exam Traps

- **Dr. Medraj Lectures 4–6 Emphasis:**
 - Step-by-step procedure for planes passing through the origin: **You must translate the origin** to an adjacent corner before identifying intercepts!
 - Close-packed stacking sequences: HCP is $ABABAB \dots$, while FCC is $ABCABCABC \dots$.
 - Single crystals are **anisotropic** (properties like Young's modulus vary with crystallographic direction), whereas polycrystalline metals with random grain orientation are **isotropic** (quasi-isotropic) on a macro scale.
- **Concordia Exam Traps:**
 - **The θ vs. 2θ Trap in XRD:** Diffractometer outputs plot intensity against 2θ (the detector angle). You **must divide by 2** before plugging θ into Bragg's Law!
 - **Volume Unit Conversion:** Forgetting that $1 \text{ nm} = 10^{-7} \text{ cm} \implies 1 \text{ nm}^3 = 10^{-21} \text{ cm}^3$.

5. 💡 Master Exam Problem & Step-by-Step Solution Framework

Problem:

Iron has a BCC crystal structure with atomic radius $R = 0.1241 \text{ nm}$ and atomic weight $A = 55.85 \text{ g/mol}$. (a) Calculate its theoretical density. (b) For monochromatic X-radiation with $\lambda = 0.1542 \text{ nm}$, compute the diffraction angle 2θ for the first-order reflection ($n = 1$) from the (220) plane.

- **Step 1: Compute BCC Lattice Parameter a**

$$a = \frac{4R}{\sqrt{3}} = \frac{4(0.1241 \text{ nm})}{\sqrt{3}} = 0.2866 \text{ nm} = 2.866 \times 10^{-8} \text{ cm}$$

- **Step 2: Compute Unit Cell Volume V_c**

$$V_c = a^3 = (2.866 \times 10^{-8} \text{ cm})^3 = 2.354 \times 10^{-23} \text{ cm}^3$$

- **Step 3: Compute Theoretical Density ρ**

$$\rho = \frac{n \cdot A}{V_c \cdot N_A} = \frac{2 \times 55.85 \text{ g/mol}}{(2.354 \times 10^{-23} \text{ cm}^3)(6.022 \times 10^{23} \text{ mol}^{-1})} = 7.88 \text{ g/cm}^3$$

- **Step 4: Compute Interplanar Spacing d_{220}**

$$d_{220} = \frac{a}{\sqrt{h^2 + k^2 + l^2}} = \frac{0.2866 \text{ nm}}{\sqrt{2^2 + 2^2 + 0^2}} = \frac{0.2866}{\sqrt{8}} = 0.1013 \text{ nm}$$

- **Step 5: Apply Bragg's Law for $\sin \theta$ and 2θ**

$$\sin \theta = \frac{n\lambda}{2d_{220}} = \frac{1(0.1542 \text{ nm})}{2(0.1013 \text{ nm})} = 0.7611$$

$$\theta = \arcsin(0.7611) = 49.56^\circ \implies 2\theta = 99.12^\circ$$

Chapter 4: Imperfections in Solids

(Aligned with Dr. Medraj Lecture 7 · Week 4)

1. 30-Second Core Intuition & Plain-English Concept

Real engineering materials are never perfect crystals. Without defects, pure metals would have theoretical shear strengths $1,000\times$ higher than observed, but they would be brittle like glass. **Defects govern all mechanical properties and atomic diffusion.**

Defects are categorized by dimensionality:

- **0D (Point Defects):** Vacancies (missing atoms), interstitials (crowded atoms), impurities.
- **1D (Linear Defects):** Dislocations (edge, screw, mixed)—the engines of plastic deformation.
- **2D (Planar Defects):** Grain boundaries, twin boundaries, external surfaces.
- **3D (Bulk Defects):** Pores, cracks, foreign inclusions.

2. High-Yield Mathematical Engine & Governing Laws

1. Equilibrium Vacancy Concentration (Arrhenius)

$$N_v = N \exp\left(-\frac{Q_v}{k_B T}\right)$$

- N_v : Number of vacancies per unit volume (extm^{-3}).
- N : Total atomic lattice sites per unit volume: $N = \frac{\rho N_A}{A}$.
- Q_v : Energy required to form a single vacancy (extJ/atom or exteV/atom).
- k_B : Boltzmann's constant ($8.62 \times 10^{-5} \text{ eV/K} = 1.38 \times 10^{-23} \text{ J/K}$).
- T : Absolute temperature in **Kelvin** ($T_K = T_C + 273.15$).

2. The Four Hume-Rothery Rules for Complete Solid Solubility

To achieve complete substitutional solubility (like Cu in Ni):

1. **Atomic Size Factor:** Difference in atomic radii must be $\Delta r = \left| \frac{r_{\text{solute}} - r_{\text{solvent}}}{r_{\text{solvent}}} \right| \leq 15$.
2. **Crystal Structure:** Must have the identical crystal structure (e.g., both FCC).
3. **Electronegativity:** Electronegativities must be very similar (large ΔX forms intermetallic compounds).
4. **Valency:** A metal dissolves more of a metal of higher valency than of lower valency.

3. Weight Percent to Atom Percent Conversion

$$C_1' = \frac{C_1/A_1}{\frac{C_1}{A_1} + \frac{C_2}{A_2}} \times 100$$

3. 🖼️ Textbook Reference Diagram & Visual Pedagogical Analysis

point defect

erties of materials are profoundly sensitive to deviations from crystalline perfection; the influence is not always adverse, and often specific characteristics are deliberately fashioned by the introduction of controlled amounts or numbers of particular defects, as detailed in succeeding chapters.

A *crystalline defect* refers to a lattice irregularity having one or more of its dimensions on the order of an atomic diameter. Classification of crystalline imperfections is frequently made according to the geometry or dimensionality of the defect. Several different imperfections are discussed in this chapter, including **point defects** (those associated with one or two atomic positions); linear (or one-dimensional) defects; and interfacial defects, or boundaries, which are two-dimensional. Impurities in solids are also discussed, because impurity atoms may exist as point defects. Finally, techniques for the microscopic examination of defects and the structure of materials are briefly described.

Point Defects

4.2 VACANCIES AND SELF-INTERSTITIALS

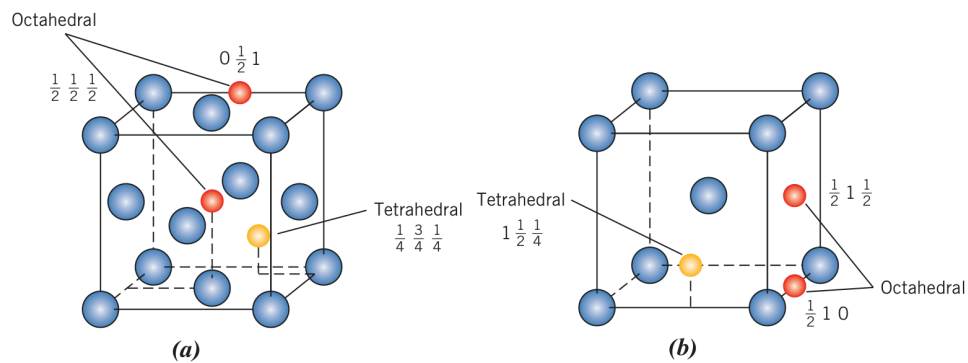
vacancy

The simplest of the point defects is a **vacancy**, or vacant lattice site, one normally occupied but from which an atom is missing (Figure 4.1). All crystalline solids contain vacancies, and, in fact, it is not possible to create such a material that is free of these

Figure 4.1: Schematic representation of a vacancy and a self-interstitial in a 2D crystalline lattice.

Figure 4.3

Locations of tetrahedral and octahedral interstitial sites within (a) FCC and (b) BCC unit cells.



have a coordination number of 4; straight lines drawn from the centers of the surrounding host atoms form a four-sided tetrahedron. However, for octahedral sites the coordination

Figure 4.3: Location of octahedral and tetrahedral interstitial voids within (a) FCC and (b) BCC crystal lattices.

In-Depth Visual Breakdown:

- **Point Defects (Figure 4.1):**
 - A **vacancy** causes surrounding lattice planes to relax inward, creating a localized tensile strain field.
 - A **self-interstitial** forces host atoms severely apart, creating a large localized compressive strain field. Because self-interstitials require massive strain energy, their equilibrium concentration is orders of magnitude lower than vacancy concentration.
- **Interstitial Sites (Figure 4.3):** Shows why carbon has vastly higher solubility in FCC austenite than in BCC ferrite. In FCC, the octahedral interstitial site at the cube edge ($[1/2, 0, 0]$) has a radius ratio of $0.414R$. In BCC, the octahedral sites on faces are distorted and much smaller ($0.154R$), severely restricting carbon solubility (0.022 wt max in BCC vs 2.14 wt in FCC).

4. 📚 Teacher Notes Cross-Reference & Concordia Exam Traps

- **Dr. Medraj Lecture 7 Focus:**

- **Arrhenius Linearization:** Taking the natural log of the vacancy equation:

$$\ln\left(\frac{N_v}{N}\right) = \ln A - \frac{Q_v}{k_B} \left(\frac{1}{T}\right)$$

Plotting $\ln(N_v/N)$ vs. $1/T$ yields a straight line with slope $= -\frac{Q_v}{k_B}$.

- **Burgers Vector Relationships:**

- **Edge dislocation:** $\vec{b} \perp \vec{t}$ (Burgers vector is perpendicular to dislocation line).
- **Screw dislocation:** $\vec{b} \parallel \vec{t}$ (Burgers vector is parallel to dislocation line).

- **Concordia Exam Traps:**

- **Temperature Units:** Forgetting to convert Celsius to Kelvin in Arrhenius calculations ($T = 800^\circ\text{C} \implies 1073.15\text{ K}$).
- **Hume-Rothery Partial Solubility Trap:** If an alloy satisfies 3 out of 4 Hume-Rothery rules (e.g., Cu-Zn: $\Delta r < 15$, similar electronegativity, but Zn is HCP while Cu is FCC), it exhibits **partial** (limited) solubility, NOT zero solubility!

5. 📌 Master Exam Problem & Step-by-Step Solution Framework

Problem:

Calculate the equilibrium number of vacancies per cubic meter in copper at 1000°C . Given: $Q_v = 0.90\text{ eV/atom}$, $\rho_{\text{Cu}} = 8.40\text{ g/cm}^3$ (at 1000°C), and $A_{\text{Cu}} = 63.55\text{ g/mol}$.

- **Step 1: Convert Temperature to Kelvin**

$$T = 1000 + 273.15 = 1273.15\text{ K}$$

- **Step 2: Calculate Number of Atomic Sites N per m^3**

$$N = \frac{\rho \cdot N_A}{A} = \frac{(8.40 \times 10^6\text{ g/m}^3)(6.022 \times 10^{23}\text{ atoms/mol})}{63.55\text{ g/mol}} = 7.96 \times 10^{28}\text{ atoms/m}^3$$

- **Step 3: Calculate the Arrhenius Factor**

$$\frac{Q_v}{k_B T} = \frac{0.90\text{ eV}}{(8.62 \times 10^{-5}\text{ eV/K})(1273.15\text{ K})} = \frac{0.90}{0.1097} = 8.204$$

$$\exp(-8.204) = 2.735 \times 10^{-4}$$

- **Step 4: Calculate Vacancy Concentration N_v**

$$N_v = N \exp\left(-\frac{Q_v}{k_B T}\right) = (7.96 \times 10^{28})(2.735 \times 10^{-4}) = 2.18 \times 10^{25}\text{ vacancies/m}^3$$

📖 Chapter 5: Diffusion

(Syllabus Week 5 · Core Midterm Topic)

1. 🎯 30-Second Core Intuition & Plain-English Concept

Diffusion is the mass transport of atoms through a solid by random atomic jumping driven by thermal vibrations. Atoms jump into neighboring vacancies (**vacancy diffusion**) or squeeze through interstices (**interstitial diffusion**).

Because small solute atoms (like C, H, N) don't need to wait for a vacancy to open up, **interstitial diffusion is thousands of times faster** than substitutional diffusion.

: Squeezing through a packed concert crowd. If you are a child (small interstitial carbon atom), you can weave between people's legs quickly. If you are a large adult (substitutional atom), you can only take a step forward when someone in front of you leaves their spot (vacancy diffusion).

2. ⚙️ High-Yield Mathematical Engine & Essential Laws

1. Fick's First Law (Steady-State Diffusion)

$$J = -D \frac{dC}{dx}$$

- J : Diffusion flux ($\text{extkg}/(\text{extm}^2 \cdot \text{s})$ or $\text{extatoms}/(\text{extm}^2 \cdot \text{s})$).
- D : Diffusion coefficient (extm^2/s).
- $\frac{dC}{dx}$: Concentration gradient (extkg/m^4). The negative sign indicates diffusion flows down the concentration gradient (from high to low concentration).

2. Fick's Second Law (Non-Steady-State Diffusion)

$$\frac{\partial C}{\partial t} = D \frac{\partial^2 C}{\partial x^2}$$

Standard Solution for Semi-Infinite Solid with Constant Surface Concentration C_s

:

$$\frac{C_x - C_0}{C_s - C_0} = 1 - \operatorname{erf}\left(\frac{x}{2\sqrt{Dt}}\right)$$

- C_0 : Uniform initial bulk concentration.
- C_s : Constant surface concentration.
- C_x : Concentration at depth x after elapsed time t .
- $\operatorname{erf}(z)$: Gaussian error function.
- **Golden Shortcut Rule**: When concentration ratio $\frac{C_x - C_0}{C_s - C_0}$ is held constant:

$$\frac{x^2}{Dt} = \text{constant} \implies \frac{x_1^2}{D_1 t_1} = \frac{x_2^2}{D_2 t_2}$$

3. Temperature Dependence of Diffusion (Arrhenius)

$$D = D_0 \exp\left(-\frac{Q_d}{RT}\right) \iff \ln D = \ln D_0 - \frac{Q_d}{R} \left(\frac{1}{T}\right)$$

- Q_d : Activation energy for diffusion (extJ/mol).
- R : Universal gas constant ($8.314 \text{ J}/\text{mol} \cdot \text{K}$).

3. 🖼️ Textbook Reference Diagram & Visual Pedagogical Analysis

(b) Note the kind of diffusion for which each of these equations is normally applied.

5.1 INTRODUCTION

diffusion

WileyPLUS

Tutorial Video:
What Is Diffusion?

interdiffusion

Many reactions and processes that are important in the treatment of materials rely on the transfer of mass either within a specific solid (ordinarily on a microscopic level) or from a liquid, a gas, or another solid phase. This is necessarily accomplished by **diffusion**, the phenomenon of material transport by atomic motion. This chapter discusses the atomic mechanisms by which diffusion occurs, the mathematics of diffusion, and the influence of temperature and diffusing species on the rate of diffusion.

The phenomenon of diffusion may be demonstrated with the use of a *diffusion couple*, which is formed by joining bars of two different metals together so that there is intimate contact between the two faces; this is illustrated for copper and nickel in Figure 5.1a, which includes schematic representations of atom positions and composition across the interface. This couple is heated for an extended period at an elevated temperature (but below the melting temperature of both metals) and cooled to room temperature. Chemical analysis reveals a condition similar to that represented in Figure 5.1b—namely, pure copper and nickel at the two extremities of the couple, separated by an alloyed region. Concentrations of both metals vary with position as shown in Figure 5.1b (bottom). This result indicates that copper atoms have migrated or diffused into the nickel, and that nickel has diffused into copper. The process by which atoms of one metal diffuse into another is termed **interdiffusion**, or **immunity diffusion**.

Figure 5.1: Copper-Nickel diffusion couple: (a) Schematic atom positions before heating, and (b) Concentration profile across the interface after elevated-temperature diffusion.

where C_x represents the concentration at depth x after time t . The expression $\text{erf}(x/2\sqrt{Dt})$ is the Gaussian error function,⁴ values of which are given in mathematical tables for various $x/2\sqrt{Dt}$ values; a partial listing is given in Table 5.1. The concentration parameters that appear in Equation 5.5 are noted in Figure 5.5, a concentration profile taken at a specific time. Equation 5.5 thus demonstrates the relationship between concentration, position, and time—namely, that C_x , being a function of the dimensionless parameter $x/\sqrt{Dt}$, may be determined at any time and position if the parameters C_0 , C_s , and D are known.

Suppose that it is desired to achieve some specific concentration of solute, C_1 , in an alloy; the left-hand side of Equation 5.5 now becomes

$$\frac{C_1 - C_0}{C_s - C_0} = \text{constant}$$

considered to be semi-infinite if none of the diffusing atoms reaches the bar end during the time t takes place. A bar of length l is considered to be semi-infinite when $l > 10\sqrt{Dt}$. The error function is defined by

$$\text{erf}(z) = \frac{2}{\sqrt{\pi}} \int_0^z e^{-y^2} dy$$

been replaced by the variable z .

Figure 5.5: Concentration profile C_x vs. depth x into a solid during gas carburizing at a specific time t .

In-Depth Visual Breakdown:

- **The Diffusion Couple (Figure 5.1):** Illustrates how an initially sharp chemical step-function (100 on left, 100 on right) gradually smooths out into an S-shaped continuous concentration curve as Cu atoms diffuse right and Ni atoms diffuse left.

- **Carburizing Profile (Figure 5.5):** Shows steel surface hardening. Carbon gas at surface maintains constant C_s . As diffusion proceeds, the carbon profile pushes deeper into the interior. The depth x at which a target hardness/carbon level is reached scales directly with $\sqrt{Dt}$.

4. 📖 Teacher Notes Cross-Reference & Concordia Exam Traps

- **Concordia Exam Focus:**
 - Gas carburizing calculations for steel gear teeth: Determining time required to achieve a specified carbon concentration at a given depth.
 - **Linear Interpolation of Error Function Table:** Concordia exams provide an abbreviated table of $\text{erf}(z)$ values; students must interpolate precisely:

$$z = z_1 + \frac{\text{erf}(z) - \text{erf}(z_1)}{\text{erf}(z_2) - \text{erf}(z_1)}(z_2 - z_1)$$

- **Concordia Exam Traps:**
 - **Gas Constant Units:** Using k_B instead of R . If Q_d is given in kJ/mol, use $R = 8.314 \text{ J/mol} \cdot \text{K}$ (multiply kJ by 1000!). If Q_d is in eV/atom, use $k_B = 8.62 \times 10^{-5} \text{ eV/K}$.

5. 💡 Master Exam Problem & Step-by-Step Solution Framework

Problem:

A gear made of 0.20 wt carbon steel is case-hardened in a gas atmosphere maintaining 1.20 wt carbon at the surface at 950°C ($D = 1.6 \times 10^{-11} \text{ m}^2/\text{s}$). How long (in hours) will it take to achieve a carbon concentration of 0.60 wt at a depth of 0.5 mm below the surface?

$$(\text{Given: } \text{erf}(0.60) = 0.6039, \text{erf}(0.65) = 0.6420)$$

- **Step 1: Set Up Fick's Second Law Concentration Ratio**

$$\frac{C_x - C_0}{C_s - C_0} = \frac{0.60 - 0.20}{1.20 - 0.20} = \frac{0.40}{1.00} = 0.40$$

- **Step 2: Solve for Error Function Value**

$$1 - \text{erf}(z) = 0.40 \implies \text{erf}(z) = 0.60$$

- **Step 3: Interpolate to Find Argument z**

$$z = 0.60 + \frac{0.6000 - 0.6039}{0.6420 - 0.6039}(0.65 - 0.60) \approx 0.595$$

- **Step 4: Relate z to Physical Diffusion Parameters**

$$z = \frac{x}{2\sqrt{Dt}} \implies 0.595 = \frac{0.5 \times 10^{-3} \text{ m}}{2\sqrt{(1.6 \times 10^{-11} \text{ m}^2/\text{s})t}}$$

$$\sqrt{t} = \frac{0.5 \times 10^{-3}}{2(0.595)\sqrt{1.6 \times 10^{-11}}} = \frac{5 \times 10^{-4}}{4.757 \times 10^{-6}} = 105.1 \text{ s}^{1/2}$$

$$t = (105.1)^2 = 11,048 \text{ seconds} = \frac{11,048}{3600} = 3.07 \text{ hours}$$

Chapter 6: Mechanical Properties of Metals

(Syllabus Week 6 · Core Midterm & Laboratory Topic)

1. 30-Second Core Intuition & Plain-English Concept

When you pull a metal rod in tension, it initially stretches like a stiff rubber band: release the load, and it snaps back to its original length (**Elastic Deformation**).

Pull it past its **Yield Strength** (σ_y), and atomic planes permanently slide past one another via dislocation slip (**Plastic Deformation**). At the peak load (**Ultimate Tensile Strength, UTS**), the cross-section necks down until fracture occurs.

Key Vocabulary Distinction

:

- **Stiffness** (E): Resistance to elastic stretching (slope of elastic region).
- **Strength** (σ_y , UTS): Resistance to permanent plastic deformation and tearing.
- **Ductility** (ϵ): Total plastic deformation sustained before snap.
- **Toughness**: Total energy absorbed up to fracture (area under entire curve).

2. High-Yield Mathematical Engine & Tensile Metrics

Property	Formula	Description & Units
Engineering Stress	$\sigma = \frac{F}{A_0}$	Force divided by <i>original</i> cross-sectional area (<i>extMPa</i> or <i>extN/mm²</i>)
Engineering Strain	$\epsilon = \frac{\Delta l}{l_0} = \frac{l - l_0}{l_0}$	Elongation divided by <i>original</i> gage length (dimensionless or <i>extmm/mm</i>)
Hooke's Law	$\sigma = E\epsilon$	Linear elastic response (E : Young's modulus, <i>extGPa</i>)
Poisson's Ratio	$\nu = -\frac{\epsilon_x}{\epsilon_z} = -\frac{\epsilon_y}{\epsilon_z}$	Lateral contraction divided by longitudinal expansion (typically ≈ 0.33)
Yield Strength (σ_y)	Stress at 0.002 (0.2) offset	Parallel line with slope E drawn starting at $\epsilon = 0.002$
Ductility (%EL)		Total permanent elongation at fracture
Modulus of Resilience	$U_r = \int_0^{\epsilon_y} \sigma d\epsilon \approx \frac{\sigma_y^2}{2E}$	Elastic energy absorbed per unit volume without permanent deformation
True Stress (σ_T)	$\sigma_T = \frac{F}{A_i} = \sigma(1 + \epsilon)$	Instantaneous force / instantaneous area (valid up to necking)
True Strain (ϵ_T)	$\epsilon_T = \ln\left(\frac{l_i}{l_0}\right) = \ln(1 + \epsilon)$	Logarithmic true strain (valid up to necking)

3. 🖼️ Textbook Reference Diagram & Visual Pedagogical Analysis

yield point, stress subsequently rises with increasing strain. For metals that display this effect, the yield strength is taken as the average stress that is associated with the lower yield point because it is well defined and relatively insensitive to the testing procedure.¹⁰ Thus, it is not necessary to employ the strain offset method for these materials.

The magnitude of the yield strength for a metal is a measure of its resistance to plastic deformation. Yield strengths may range from 35 MPa (5000 psi) for a low-strength aluminum to greater than 1400 MPa (200,000 psi) for high-strength steels.

Concept Check 6.1 Cite the primary differences between elastic, anelastic, and plastic deformation behaviors.

The answer may be found in all digital versions of the text and/or at www.wiley.com/college/callister (Student Companion Site).]

Tensile Strength

After yielding, the stress necessary to continue plastic deformation in metals increases to a maximum, point M in Figure 6.11, and then decreases to the eventual fracture, point F . The **tensile strength** TS (MPa or psi) is the stress at the maximum on the engineering stress-strain curve (Figure 6.11). This corresponds to the maximum stress that can be sustained by a structure in tension; if this stress is applied and maintained, fracture will result. All deformation to this point is uniform throughout the narrow region of the tensile specimen. However, at this maximum stress, a small constriction or neck begins to form at some point, and all subsequent deformation is confined at this neck, as indicated by the schematic specimen insets in Figure 6.11. This phenomenon is termed *necking*,

in lieu of *stress* because strength is a property of the metal, whereas stress is related to the magnitude of the force. In the United States, the unit of kilopounds per square inch (ksi) is sometimes used for the sake of convenience, where

Figure 6.11: Engineering stress-strain behavior to fracture: P (proportional limit), M (maximum load / tensile strength), and F (fracture point).

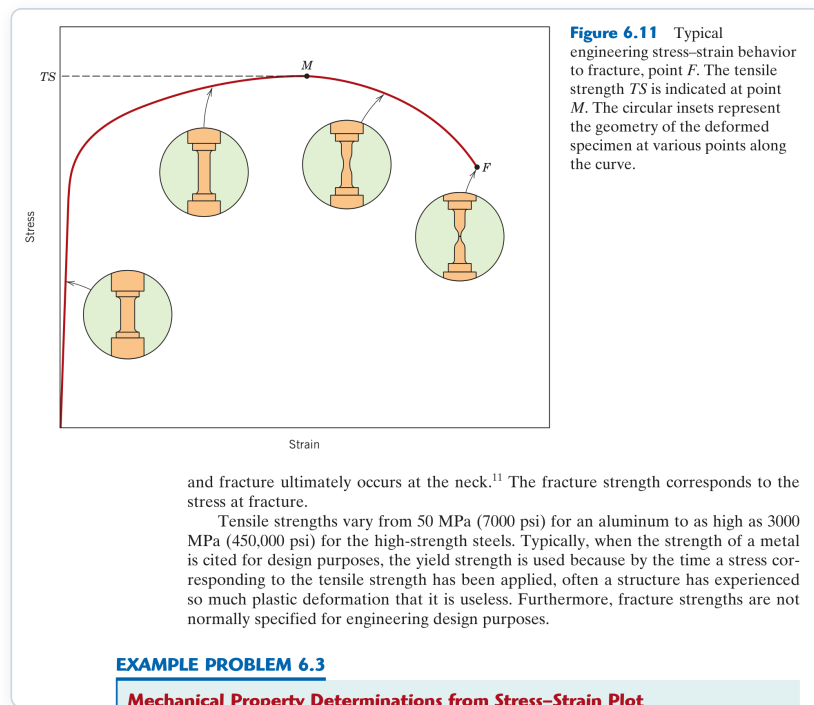


Figure 6.12: Determination of the 0.002 (0.2) strain offset yield strength for brass.

In-Depth Visual Breakdown:

- **The Engineering Curve (Figure 6.11):**
 - The curve appears to drop after peak M (Tensile Strength) because engineering stress uses *original area*

A_0 . In reality, the material continues to strain harden, but localized **necking** causes the cross-section to shrink faster than the load increases.

- **The 0.002 Offset Method (Figure 6.12):**

- For metals without a sharp yield drop (like brass, aluminum, copper), yield strength is standardized by starting at $\epsilon = 0.002$ (0.2) on the horizontal axis and drawing a dashed line strictly **parallel** to the initial linear elastic slope E . The intersection with the experimental curve defines σ_y .

4. 🏛️ Teacher Notes Cross-Reference & Concordia Exam Traps

- **Concordia Exam & Lab Focus:**

- Calculating Young's modulus, yield strength (0.2), UTS, and ductility directly from raw load-elongation tensile data.
- Modulus of resilience calculation: $U_r = \frac{\sigma_y^2}{2E}$.
- Hardness correlations: For steel, $\text{UTS (MPa)} \approx 3.45 \times \text{HB}$ (Brinell Hardness).

- **Concordia Exam Traps:**

- **True vs. Engineering Equations:** The equations $\sigma_T = \sigma(1 + \epsilon)$ and $\epsilon_T = \ln(1 + \epsilon)$ are **ONLY valid up to the onset of necking (UTS)**! After necking initiates, cross-sectional deformation is no longer uniform, and instantaneous cross-sectional area must be measured directly.

5. 💡 Master Exam Problem & Step-by-Step Solution Framework

Problem:

A cylindrical specimen of aluminum alloy with diameter $d_0 = 12.8 \text{ mm}$ and gage length $l_0 = 50.8 \text{ mm}$ is loaded in tension. An elastic load of $15,000 \text{ N}$ produces an elongation of 0.089 mm . The specimen yields at $30,000 \text{ N}$, reaches a maximum load of $45,000 \text{ N}$, and fractures at $l_f = 56.4 \text{ mm}$. Calculate: (a) Young's Modulus E , (b) Yield Strength σ_y , (c) Tensile Strength UTS, (d) Ductility, and (e) Modulus of Resilience U_r .

- **Step 1: Compute Initial Area A_0**

$$A_0 = \frac{\pi d_0^2}{4} = \frac{\pi (12.8 \text{ mm})^2}{4} = 128.7 \text{ mm}^2 = 1.287 \times 10^{-4} \text{ m}^2$$

- **Step 2: Compute Modulus of Elasticity E**

$$\sigma_1 = \frac{15,000 \text{ N}}{128.7 \text{ mm}^2} = 116.55 \text{ MPa}, \quad \epsilon_1 = \frac{0.089 \text{ mm}}{50.8 \text{ mm}} = 0.001752$$

$$E = \frac{\sigma_1}{\epsilon_1} = \frac{116.55 \text{ MPa}}{0.001752} = 66,524 \text{ MPa} = 66.5 \text{ GPa}$$

- **Step 3: Compute Yield Strength σ_y and Tensile Strength UTS**

$$\sigma_y = \frac{F_{\text{yield}}}{A_0} = \frac{30,000 \text{ N}}{128.7 \text{ mm}^2} = 233.1 \text{ MPa}$$

$$\text{UTS} = \frac{F_{\text{max}}}{A_0} = \frac{45,000 \text{ N}}{128.7 \text{ mm}^2} = 349.6 \text{ MPa}$$

- **Step 4: Compute Ductility**

- **Step 5: Compute Modulus of Resilience U_r**

$$U_r \approx \frac{\sigma_y^2}{2E} = \frac{(233.1 \times 10^6 \text{ Pa})^2}{2(66.5 \times 10^9 \text{ Pa})} = 4.08 \times 10^5 \text{ J/m}^3 = 0.408 \text{ MJ/m}^3$$

Chapter 7: Dislocations and Strengthening Mechanisms

(Syllabus Week 7 · Sections 7.1–7.4 · Midterm & Final Scope)

1. 30-Second Core Intuition & Plain-English Concept

Plastic deformation in metals occurs when line defects (dislocations) slide across close-packed atomic planes.

The Universal Law of Metallurgy: To make a metal stronger, you must **impede dislocation motion**. If dislocations cannot move, the metal cannot permanently deform—its yield strength skyrockets!

There are **Four Universal Strengthening Mechanisms**:

1. **Grain Size Reduction:** Grain boundaries block dislocations.
2. **Solid Solution Strengthening:** Solute atoms create lattice strain fields that pin dislocations.
3. **Strain Hardening (Cold Work):** Tangling dislocations into each other.
4. **Precipitation Hardening:** Hard microscopic particles act as barricades.

2. High-Yield Mathematical Engine & Governing Laws

1. Schmid's Law for Resolved Shear Stress

$$\tau_R = \sigma \cos \phi \cos \lambda$$

- σ : Applied macroscopic tensile stress.
- ϕ : Angle between tensile axis and the **normal to the slip plane**.
- λ : Angle between tensile axis and the **slip direction**.
- $(\cos \phi \cos \lambda)$: **Schmid Factor (m)**. Maximum theoretical value is 0.5 (when $\phi = \lambda = 45^\circ$).
- Yielding initiates in a single crystal when $\tau_R = \tau_{\text{crss}}$ (Critical Resolved Shear Stress):

$$\sigma_y = \frac{\tau_{\text{crss}}}{(\cos \phi \cos \lambda)_{\text{max}}}$$

2. Hall-Petch Relationship (Grain Size Reduction)

$$\sigma_y = \sigma_0 + k_y d^{-1/2}$$

- σ_0 : Friction stress resisting dislocation motion in a single crystal.
- k_y : Strengthening coefficient (material constant).
- d : Average grain diameter. Smaller grain size $d \implies$ higher yield strength σ_y .
- *Unique Advantage*

: Grain size reduction is the **only strengthening mechanism that increases both strength AND toughness simultaneously!**

3. Strain Hardening (Cold Work)

- Flow stress in plastic region: $\sigma_T = K \epsilon_T^n$ (n : strain-hardening exponent).

3. 🖼️ Textbook Reference Diagram & Visual Pedagogical Analysis

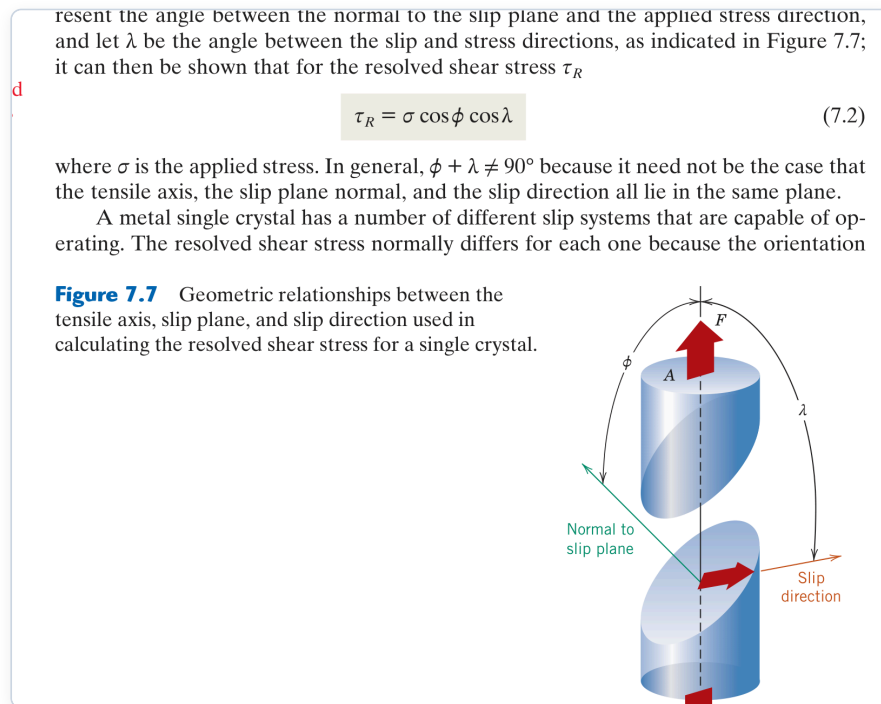


Figure 7.7: Geometric relationships between tensile axis, slip plane normal (ϕ), and slip direction (λ) in a single crystal.

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hardness and strength (both yield and tensile) are related to the ease with which plastic deformation can be made to occur, by reducing the mobility of dislocations, the mechanical strength may be enhanced—that is, greater mechanical forces are required to initiate plastic deformation. In contrast, the more unconstrained the dislocation motion, the greater is the facility with which a metal may deform, and the softer and weaker it becomes. Virtually all strengthening techniques rely on this simple principle: *Restricting or hindering dislocation motion renders a material harder and stronger.*

The present discussion is confined to strengthening mechanisms for single-phase metals by grain size reduction, solid-solution alloying, and strain hardening. Deformation and strengthening of multiphase alloys are more complicated, involving concepts beyond the scope of the present discussion; Chapter 10 and Section 11.10 treat techniques that are used to strengthen multiphase alloys.

STRENGTHENING BY GRAIN SIZE REDUCTION

The size of the grains, or average grain diameter, in a polycrystalline metal influences the mechanical properties. Adjacent grains normally have different crystallographic orientations and, of course, a common grain boundary, as indicated in Figure 7.14. During plastic deformation, slip or dislocation motion must take place across this common boundary—say, from grain A to grain B in Figure 7.14. The grain boundary acts as a barrier to dislocation motion for two reasons:

Figure 7.14: Dislocation pile-up at a grain boundary. The atomic mismatch between adjacent grains blocks dislocation slip.

In-Depth Visual Breakdown:

- **Schmid's Law Geometry (Figure 7.7):** Demonstrates that tension does not directly cause slip; rather, slip is driven by the **shear component** resolved onto the slip plane along the slip direction. If the tensile axis is perpendicular to the slip plane ($\phi = 0^\circ \implies \lambda = 90^\circ$), the resolved shear stress is zero ($\cos 90^\circ = 0$), and no slip can occur!

- **Hall-Petch Barrier (Figure 7.14):** When dislocations move along slip plane A in Grain 1, they slam into the grain boundary. Because Grain 2 has a different crystallographic orientation, slip plane B is misaligned. Dislocations pile up at the boundary, generating a back-stress that resists further deformation until applied stress is raised substantially.

4. 🏛️ Teacher Notes Cross-Reference & Concordia Exam Traps

- **Concordia Exam Focus:**
 - Fitting two data points to the Hall-Petch equation to find σ_0 and k_y , then predicting σ_y for a third grain size.
 - Calculating and reading resulting yield strength, tensile strength, and ductility from standard empirical curves.
 - The three annealing stages: **Recovery** (internal stresses relieve, conductivity restored), **Recrystallization** (new strain-free grains nucleate, ductility restored, strength drops), and **Grain Growth** (grains coarsen to reduce boundary energy).
- **Concordia Exam Traps:**
 - **Schmid's Law Angle Trap:** In 3D space, $\phi + \lambda \neq 90^\circ$! They are independent angles measured between the tensile axis and two separate vectors. Do not assume $\cos \lambda = \sin \phi$.

5. 💡 Master Exam Problem & Step-by-Step Solution Framework

Problem:

A metal with average grain diameter $d_1 = 0.050 \text{ mm}$ has a yield strength of 135 MPa . When grain size is reduced to $d_2 = 0.008 \text{ mm}$, yield strength increases to 260 MPa . Calculate: (a) The Hall-Petch constants σ_0 and k_y , and (b) The expected yield strength when average grain diameter is 0.002 mm .

- **Step 1: Compute $d^{-1/2}$ for Known Grain Sizes**

$$d_1^{-1/2} = (0.050 \text{ mm})^{-1/2} = 4.472 \text{ mm}^{-1/2}$$

$$d_2^{-1/2} = (0.008 \text{ mm})^{-1/2} = 11.180 \text{ mm}^{-1/2}$$

- **Step 2: Set Up Simultaneous Linear Equations**

$$135 = \sigma_0 + k_y(4.472) \quad \text{--- (1)}$$

$$260 = \sigma_0 + k_y(11.180) \quad \text{--- (2)}$$

- **Step 3: Solve for k_y and σ_0**

$$\text{Subtract (1) from (2) : } 125 = k_y(11.180 - 4.472) = 6.708k_y$$

$$k_y = \frac{125}{6.708} = 18.63 \text{ MPa} \cdot \text{mm}^{1/2}$$

$$\sigma_0 = 135 - (18.63)(4.472) = 135 - 83.31 = 51.69 \text{ MPa}$$

- **Step 4: Predict Yield Strength for $d = 0.002 \text{ mm}$**

$$d_3^{-1/2} = (0.002 \text{ mm})^{-1/2} = 22.361 \text{ mm}^{-1/2}$$

$$\sigma_y = 51.69 + (18.63)(22.361) = 51.69 + 416.58 = 468.3 \text{ MPa}$$

Chapter 9: Phase Diagrams & Microstructural Evolution

1. 🌀 30-Second Core Intuition & Plain-English Concept

A phase diagram is a metallurgical roadmap showing what microscopic phases exist at any combination of temperature and chemical composition.

From a phase diagram, you can answer three vital questions for any alloy:

1. **What phases are present?** (Look at which field the point lands in).
2. **What is the chemical composition of each phase?** (Draw a horizontal **tie-line** and read the intersections).
3. **How much of each phase is present?** (Apply the **Inverse Lever Rule**).

Real-World Analogy

: Making chocolate milk. Below the solubility limit, cocoa dissolves completely into milk (single liquid phase). Add too much cocoa powder, and solid sludge settles at the bottom: now you have two phases (saturated liquid milk + solid cocoa powder) in equilibrium.

2. ⚙️ High-Yield Mathematical Engine & The Lever Rule

1. Gibbs Phase Rule (Condensed System at 1extatm)

$$P + F = C + 1$$

- P : Number of phases present.
- F : Degrees of freedom (number of externally controllable variables: T , composition).
- C : Number of chemical components (e.g., $C = 2$ for binary systems like Cu-Ni or Fe-C).

2. The Inverse Lever Rule (Phase Fraction Calculation)

In a two-phase region ($\alpha + L$) with overall alloy composition C_0 :

$$W_L = \frac{C_\alpha - C_0}{C_\alpha - C_L} \quad (\text{Fraction of Liquid})$$

$$W_\alpha = \frac{C_0 - C_L}{C_\alpha - C_L} \quad (\text{Fraction of Solid } \alpha)$$

(Notice the inverse nature: the amount of α is proportional to the lever arm on the liquid side!)

3. Core Invariant Reactions

- **Eutectic:** $L \xrightarrow{\text{cooling}} \alpha + \beta$ (Liquid freezes into two intimate solid phases).
- **Eutectoid:** $\gamma \xrightarrow{\text{cooling}} \alpha + \beta$ (Solid transforms into two new solid phases).
- **Peritectic:** $L + \alpha \xrightarrow{\text{cooling}} \beta$.

4. The Iron-Carbon ($Fe - Fe_3C$) System (The Heart of Metallurgy)

- **Ferrite (α):** BCC iron. Stable at room temperature. Extremely low carbon solubility (max 0.022 wt at 727°C). Soft and ductile.
- **Austenite (γ):** FCC iron. Stable between 912°C and 1394°C . High carbon solubility (max 2.14 wt at 1147°C).
- **Cementite (Fe_3C):** Stoichiometric iron carbide containing 6.70wt% **Carbon**. Extremely hard and brittle.
- **Eutectoid Reaction at 727°C and 0.76wt%:**

$$\gamma(0.76$$

: Alternating microscopic lamellae (plates) of soft ferrite and hard cementite.

3. 🖼️ Textbook Reference Diagram & Visual Pedagogical Analysis

with binary alloys—those that contain two components. If more than two components are present, phase diagrams become extremely complicated and difficult to represent. An explanation of the principles governing and the interpretation of phase diagrams can be demonstrated using binary alloys even though most alloys contain more than two components.

Binary phase diagrams are maps that represent the relationships between temperature and the compositions and quantities of phases at equilibrium, which influence the microstructure of an alloy. Many microstructures develop from *phase transformations*, the changes that occur when the temperature is altered (typically upon cooling). This may involve the transition from one phase to another or the appearance or disappearance of a phase. Binary phase diagrams are helpful in predicting phase transformations and the resulting microstructures, which may have equilibrium or nonequilibrium character.

ISOMORPHOUS SYSTEMS

Possibly the easiest type of binary phase diagram to understand and interpret is the type that is characterized by the copper–nickel system (Figure 9.3a). Temperature is plotted along the ordinate, and the abscissa represents the composition of the alloy, in weight percent of nickel. The composition ranges from 0 wt% Ni (100 wt% Cu) on the far left horizontal extreme to 100 wt% Ni (0 wt% Cu) on the right. Three different phase regions, or *fields*, appear on the diagram: an alpha (α) field, a liquid (*L*) field, and a two-phase $\alpha + L$ field. Each region is defined by the phase or phases that exist over the range of temperatures and compositions delineated by the phase boundary lines.

Figure 9.3: Tie-line and lever rule construction in a binary isomorphous system.

9.18 THE IRON–IRON CARBIDE ($\text{Fe}-\text{Fe}_3\text{C}$) PHASE DIAGRAM

ferrite
austenite

A portion of the iron–carbon phase diagram is presented in Figure 9.24. Pure iron, upon heating, experiences two changes in crystal structure before it melts. At room temperature, the stable form, called **ferrite**, or α -iron, has a BCC crystal structure. Ferrite experiences a polymorphic transformation to FCC **austenite**, or γ -iron, at 912°C (1674°F). This austenite persists to 1394°C (2541°F), at which temperature the FCC austenite reverts back to a BCC phase known as δ -ferrite, which finally melts

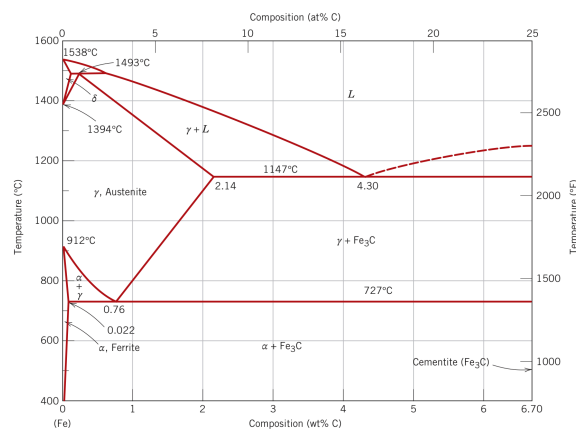


Figure 9.24 The iron–iron carbide phase diagram.
[Adapted from *Binary Alloy Phase Diagrams*, Vol. 1, 2nd edition, T. B. Massalski (Editor-in-Chief), 1990. Reprinted by permission of ASM International, Materials Park, OH.]

Figure 9.24: The Iron-Iron Carbide ($\text{Fe} - \text{Fe}_3\text{C}$) phase diagram showing ferrite (α), austenite (γ), and cementite (Fe_3C).

In-Depth Visual Breakdown:

- **The Lever Rule (Figure 9.3):** At point *B* (temperature T_0 , composition C_0):

- Draw horizontal tie-line from solidus line C_{α} to liquidus line C_L .
- The total tie-line length is $(C_{\alpha} - C_L)$.
- The weight fraction of liquid W_L is the length of the segment opposite to the liquidus line, divided by total length: $W_L = \frac{C_{\alpha} - C_0}{C_{\alpha} - C_L}$.
- **The $Fe - Fe_3C$ Phase Diagram (Figure 9.24):** The cornerstone of mechanical engineering materials:
 - **Steels** contain < 2.14 wt (typically $0.05 - 1.2$ wt).
 - **Cast Irons** contain > 2.14 wt (typically $3.0 - 4.5$ wt).
 - **Hypoeutectoid Steels** ($C_0 < 0.76$): Cool to form proeutectoid ferrite + pearlite.
 - **Hypereutectoid Steels** ($C_0 > 0.76$): Cool to form proeutectoid cementite + pearlite.

4. Teacher Notes Cross-Reference & Concordia Exam Traps

- **Concordia Exam Focus:**
 - Calculating the fraction of **proeutectoid ferrite** vs. **eutectoid ferrite** vs. **total ferrite** in hypoeutectoid steels.
- **The Classic Concordia Exam Trap:**
 - **Total Ferrite vs. Proeutectoid Ferrite:**
 - **Total Ferrite (α_{total})** is calculated across the entire baseline tie-line at 726°C (0.022 to 6.70):

$$W_{\alpha,\text{total}} = \frac{6.70 - C_0}{6.70 - 0.022}$$

- **Proeutectoid Ferrite (α_{pro})** is calculated
just above the eutectoid temperature

(728°C) between 0.022 and 0.76:

$$W_{\alpha,\text{pro}} = \frac{0.76 - C_0}{0.76 - 0.022}$$

- **Eutectoid Ferrite** is the ferrite residing
inside

the pearlite colonies:

$$W_{\alpha,\text{eutectoid}} = W_{\alpha,\text{total}} - W_{\alpha,\text{pro}}$$

5. Master Exam Problem & Step-by-Step Solution Framework

Problem (Classic Concordia Final Exam Question):

For a $0.35\text{ wt}\%$ plain carbon steel cooled slowly to just below 727°C , calculate: (a) The mass fraction of total ferrite (α) and total cementite (Fe_3C), and (b) The mass fraction of proeutectoid ferrite and pearlite.

- **Step 1: Compute Total Phases Just Below 727°C (Tie-line: 0.022 to 6.70)**

$$W_{\alpha,\text{total}} = \frac{6.70 - 0.35}{6.70 - 0.022} = \frac{6.35}{6.678} = 0.951 \quad (95.1)$$

$$W_{Fe_3C,\text{total}} = \frac{0.35 - 0.022}{6.70 - 0.022} = \frac{0.328}{6.678} = 0.049 \quad (4.9)$$

- **Step 2: Compute Microconstituents Just Above 727°C (Tie-line: 0.022 to 0.76)**

At 728°C , the steel consists of proeutectoid α and austenite γ

:

$$W_{\alpha,\text{pro}} = \frac{0.76 - 0.35}{0.76 - 0.022} = \frac{0.41}{0.738} = 0.556 \quad (55.6)$$

$$W_{\gamma} = \frac{0.35 - 0.022}{0.76 - 0.022} = \frac{0.328}{0.738} = 0.444 \quad (44.4)$$

- **Step 3: Relate Austenite to Pearlite**

Upon cooling through 727°C, all remaining austenite of composition 0.76 transforms 1-to-1 into pearlite

:

$$W_{\text{pearlite}} = W_{\gamma} = 0.444 \quad (44.4)$$

- **Step 4: Verify Consistency**

$$W_{\alpha,\text{pro}} + W_{\text{pearlite}} = 0.556 + 0.444 = 1.000 \quad (100)$$

Amount of eutectoid ferrite inside pearlite

:

$$W_{\alpha,\text{eutectoid}} = 0.951 - 0.556 = 0.395 \quad (39.5)$$

Chapters 12 & 13: Structures, Properties and Processing of Ceramics

(Syllabus Week 9 · Sections 12.3–12.11 & 13.11)

1. 30-Second Core Intuition & Plain-English Concept

Ceramics are compounds of metallic and nonmetallic elements held by strong **ionic and covalent bonds**.

Because ions have fixed positive and negative charges, like-charges fiercely repel if atomic planes attempt to slide. Therefore, **dislocation motion is virtually impossible** at room temperature. Ceramics cannot plastically deform; under tension, microscopic surface microcracks concentrate stress and propagate instantaneously at the speed of sound (**brittle catastrophic fracture**).

Real-World Analogy

: Trying to slide a row of alternating north-south magnets past another row. As soon as you shift them half a notch, all north poles align with north poles, and the entire structure blasts violently apart.

2. High-Yield Mathematical Engine & Ceramic Geometry

1. Cation-to-Anion Radius Ratio (r_C/r_A) Coordination Table

Cations are smaller than anions ($r_C < r_A$). Stable crystal structures require cations to maximize contact with surrounding anions without allowing anions to overlap:

Radius Ratio (r_C/r_A)	Coordination Number (CN)	Coordination Geometry	Archetypal Ceramic Crystal Structure
< 0.155	2	Linear	Inert gas configurations
$0.155 - 0.225$	3	Triangular planar	$extB_2extO_3$
$0.225 - 0.414$	4	Tetrahedral	Zinc Blende ($extZnS$), $extSiO_4^{4-}$
$0.414 - 0.732$	6	Octahedral	Rock Salt ($extNaCl$, $extMgO$, $extFeO$)
$0.732 - 1.000$	8	Cubic	Cesium Chloride ($extCsCl$)
> 1.000	12	Close-packed	Rare intermetallic ceramics

2. Defect Chemistry in Ceramics

To preserve **electroneutrality** (charge neutrality):

- **Frenkel Defect:** A cation vacates its normal lattice site and squeezes into a nearby interstitial position (cation vacancy + cation interstitial; no net charge change).
- **Schottky Defect:** A stoichiometric pair consisting of **one cation vacancy AND one anion vacancy** (e.g., in $extNaCl$, one $extNa^+$ missing and one $extCl^-$ missing; net charge remains zero).

3. Flexural Strength via Three-Point Bending (σ_{fs})

Because ceramics break prematurely in tension grips, tensile strength is measured via transverse bending:

$$\sigma_{fs} = \frac{3F_f L}{2bd^2} \quad (\text{Rectangular cross-section: width } b, \text{ depth } d, \text{ support span } L)$$

$$\sigma_{fs} = \frac{F_f L}{\pi R^3} \quad (\text{Circular cross-section of radius } R)$$

3. Textbook Reference Diagram & Visual Pedagogical Analysis

$$\overline{AO} = r_A + r_C$$

Furthermore, the side length ratio $\overline{AP}/\overline{AO}$ is a function of the angle α as

$$\frac{\overline{AP}}{\overline{AO}} = \cos \alpha$$

The magnitude of α is 30° , because line $\overline{AO}$ bisects the 60° angle BAC . Thus,

$$\frac{\overline{AP}}{\overline{AO}} = \frac{r_A}{r_A + r_C} = \cos 30^\circ = \frac{\sqrt{3}}{2}$$

Solving for the cation–anion radius ratio,

$$\frac{r_C}{r_A} = \frac{1 - \sqrt{3}/2}{\sqrt{3}/2} = 0.155$$

AX-Type Crystal Structures

Some of the common ceramic materials are those in which there are equal numbers of cations and anions. These are often referred to as AX compounds, where A denotes the cation and X the anion. There are several different crystal structures for AX compounds; each is typically named after a common material that assumes the particular structure.

Figure 12.2: Rock Salt (NaCl) unit cell. FCC anion packing with cations occupying all octahedral interstices.

oxides do not form polyhedral networks. Rather, their cations are incorporated within and modify the SiO_4^{4-} network; for this reason, these oxide additives are termed *network modifiers*. For example, Figure 12.11 is a schematic representation of the structure of a sodium–silicate glass. Still other oxides, such as TiO_2 and Al_2O_3 although not network formers, substitute for silicon and become part of and stabilize the network; these are called *intermediates*. From a practical perspective, the addition of these modifiers and intermediates lowers the melting point and viscosity of a glass and makes it easier to form at lower temperatures (Section 13.11).

The Silicates

For the various silicate minerals, one, two, or three of the corner oxygen atoms of the SiO_4^{4-} tetrahedra are shared by other tetrahedra to form some rather complex structures. Some of these, represented in Figure 12.12, have formulas SiO_4^{4-} , $\text{Si}_2\text{O}_7^{6-}$, $\text{Si}_3\text{O}_9^{6-}$ and so on; single-chain structures are also possible, as in Figure 12.12e. Positively charged cations such as Ca^{2+} , Mg^{2+} , and Al^{3+} serve two roles: First, they compensate the negative charges from the SiO_4^{4-} units so that charge neutrality is achieved; second, these cations ionically bond the SiO_4^{4-} tetrahedra together.

Figure 12.10: 2D representation of noncrystalline silica glass with network modifier cations (Na^+ , Ca^{2+}) disrupting the bridging oxygen network.

raised, the magnitude of the bonding is diminished, the sliding motion or flow of the atoms or ions is facilitated, and subsequently there is an attendant decrease in viscosity. A discussion of the temperature dependence of viscosity for glasses is deferred to Section 13.11.

Figure 12.32 Representation of the viscous flow of a liquid or fluid glass in response to an applied shear force.

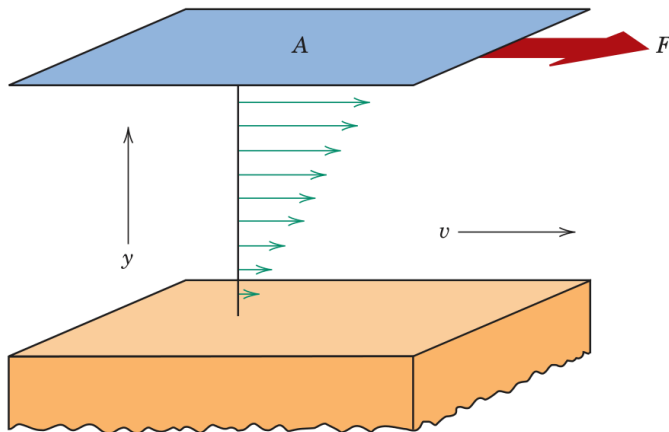


Figure 12.32: Three-point bending fixture for measuring flexural strength of brittle ceramics.

In-Depth Visual Breakdown:

- **Rock Salt Structure (Figure 12.2):** Anions (Cl^-) form an FCC unit cell lattice. Octahedral interstitial sites are located at the center of the unit cell and at the midpoints of all 12 cube edges. Cations (Na^+) fill all octahedral sites ($n = 4$ formula units per cell).
- **Silicate Glass (Figure 12.10):** Pure crystalline silica (SiO_2) forms a rigid, ordered network of corner-sharing SiO_4^{4-} tetrahedra. Adding network modifiers like soda (Na_2O) and lime (CaO) introduces non-

bridging oxygen atoms, breaking up network connectivity, drastically lowering the melting point from 1710°C to $\approx 1000^{\circ}\text{C}$ to produce workable window glass.

4. 📚 Teacher Notes Cross-Reference & Concordia Exam Traps

- **Concordia Exam Focus:**
 - Predicting ceramic coordination number and crystal structure from ion radii (r_{cation} and r_{anion}).
 - Calculating flexural strength σ_{fs} from 3-point bend test failure loads.
 - Explaining why ceramics have high compressive strength (10 times higher) but low tensile strength (Griffith flaw theory: microcracks open in tension but close harmlessly in compression).
- **Concordia Exam Traps:**
 - Forgetting that in the 3-point bending formula for rectangular bars ($\sigma_{fs} = \frac{3FL}{2bd^2}$), the depth d is **squared**, whereas width b is linear. Inverting b and d will produce massive numerical errors!

5. 💡 Master Exam Problem & Step-by-Step Solution Framework

Problem:

Magnesium oxide (MgO) has $r_{\text{Mg}^{2+}} = 0.072\text{ nm}$ and $r_{\text{O}^{2-}} = 0.140\text{ nm}$. (a) Predict the coordination number and crystal structure. (b) A rectangular bar of MgO ($b = 10\text{ mm}$, $d = 5\text{ mm}$) tested in a 3-point bend fixture with span $L = 50\text{ mm}$ fractures at load $F_f = 2,500\text{ N}$. Compute its flexural strength.

- **Step 1: Compute Cation-Anion Radius Ratio**

$$\frac{r_{\text{Mg}^{2+}}}{r_{\text{O}^{2-}}} = \frac{0.072\text{ nm}}{0.140\text{ nm}} = 0.514$$

- **Step 2: Predict Coordination and Structure**

Since $0.414 < 0.514 < 0.732$, the coordination number is **6 (Octahedral)**. Because stoichiometry is 1 : 1 (AX type), MgO crystallizes in the **Rock Salt (NaCl) crystal structure**.

- **Step 3: Calculate Flexural Strength σ_{fs}**

$$\sigma_{fs} = \frac{3F_f L}{2bd^2} = \frac{3(2,500\text{ N})(0.050\text{ m})}{2(0.010\text{ m})(0.005\text{ m})^2} = \frac{375}{2(0.010)(2.5 \times 10^{-5})} = \frac{375}{5.0 \times 10^{-7}} = 750 \times 10^6\text{ Pa} = 750\text{ MPa}$$

📖 Chapters 14 & 15: Polymer Structures and Properties

(Syllabus Week 10 · Sections 14.1–14.15 & 15.1–15.14)

1. 🎯 30-Second Core Intuition & Plain-English Concept

Polymers are gigantic macromolecular chains made of repeating chemical units (monomers) linked by strong covalent bonds.

The mechanical behavior depends entirely on whether chains are free to slide past one another (**Thermoplastics**) or permanently locked together into an un-meltable 3D net (**Thermosets**).

Key Real-World Distinction

:

- **Thermoplastic** (e.g., Polyethylene, Nylon): Like a bowl of cooked spaghetti. When heated, chains slide freely and the plastic melts. Can be reheated and recycled indefinitely.

- **Thermoset** (e.g., Epoxy, Vulcanized Rubber): Like a baked cake. Chemical cross-links form permanent covalent bridges across all chains. Reheating will not melt it; it will simply char and burn. Cannot be recycled by melting.

2. ⚙️ High-Yield Mathematical Engine & Molecular Metrics

1. Molecular Weight and Degree of Polymerization

Polymer chains have varying lengths, described by statistical distributions:

- **Number-Average Molecular Weight:** $\bar{M}_n = \sum x_i M_i$ (x_i : number fraction).
- **Weight-Average Molecular Weight:** $\bar{M}_w = \sum w_i M_i$ (w_i : weight fraction).
- **Polydispersity Index (PDI):** $PDI = \frac{\bar{M}_w}{\bar{M}_n} \geq 1.0$ (Measures breadth of chain length distribution).
- **Degree of Polymerization (DP):**

$$DP = \frac{\bar{M}_n}{m}$$

where m is the molecular weight of the single repeating monomer unit.

2. Molecular Architectures & Tacticity

- **Architectures:** Linear $\rightarrow$ Branched $\rightarrow$ Cross-linked $\rightarrow$ Network.
- **Tacticity (Stereoisomerism):**
 - **Isotactic:** All side R -groups positioned on the *same side* of the chain (crystallizes easily).
 - **Syndiotactic:** R -groups alternate regularly from *side to side*.
 - **Atactic:** R -groups positioned *randomly* (prevents chain packing; completely amorphous).

3. Thermal Transitions

- **Glass Transition Temperature (T_g):** Temperature below which amorphous polymer chains freeze into a brittle, glassy state; above T_g , chains become mobile, leathery, and rubbery.
- **Melting Temperature (T_m):** Temperature where crystalline ordered lamellae melt into a disordered liquid.

3. Textbook Reference Diagram & Visual Pedagogical Analysis

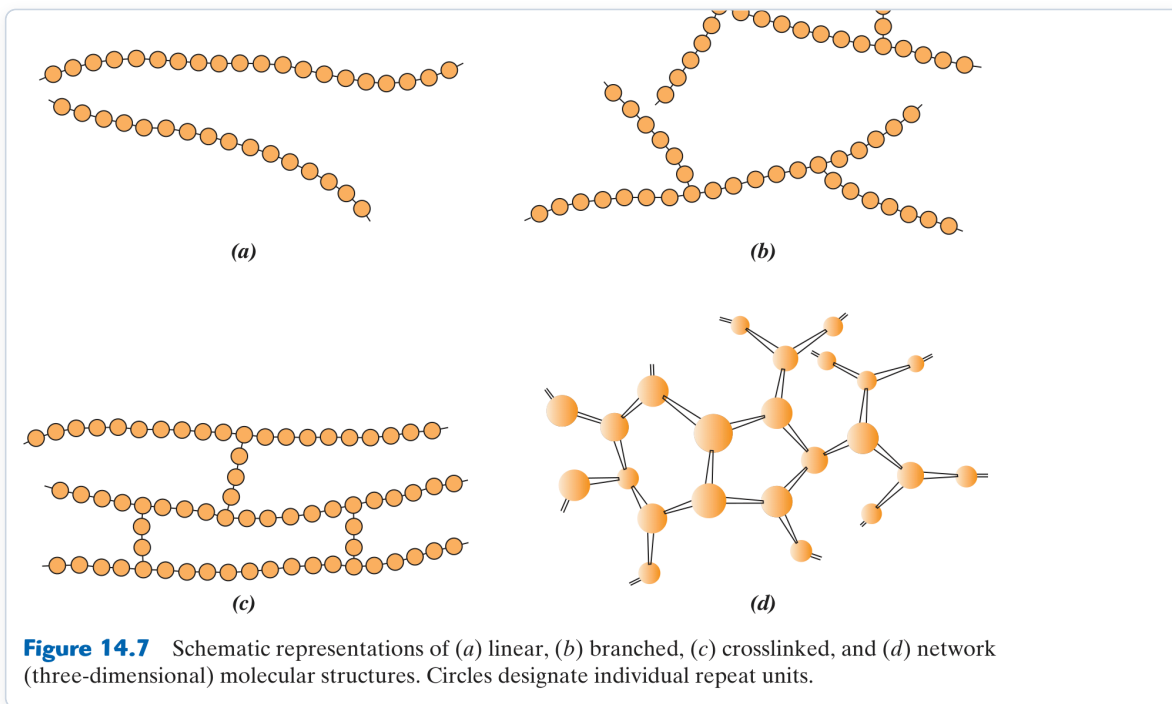


Figure 14.7: Schematic representation of polymer chain structures: (a) Linear, (b) Branched, (c) Crosslinked, and (d) Network.

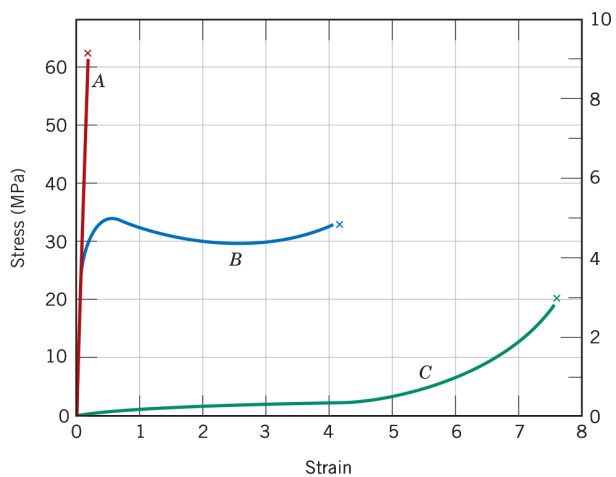


Figure 15.1 The stress-strain behavior for brittle (curve A), plastic (curve B), and highly elastic (elastomeric) (curve C) polymers.

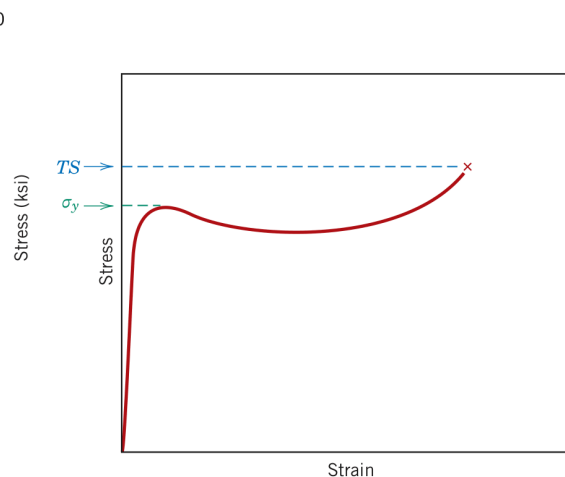


Figure 15.2 Schematic stress-strain curve for a plastic polymer showing how yield and tensile strengths are determined.

Figure 15.1: Stress-strain behavior for polymers: Curve A (Brittle polymer below T_g), Curve B (Plastic polymer with necking/drawing), and Curve C (Highly elastic elastomer).

In-Depth Visual Breakdown:

- **Chain Architectures (Figure 14.7):**

- (a) Linear
: High packing density (HDPE); high crystallinity and strength.
- (b) Branched

: Side branches prevent tight packing (LDPE); lower density and greater flexibility.

- o (c) Crosslinked

: Adjacent chains joined by covalent bonds (vulcanized rubber with sulfur crosslinks); imparts elasticity.

- o (d) Network

: Fully 3D covalent crosslinking (epoxies, phenolics); rigid and completely non-meltable.

- **Polymer Stress-Strain Curves (Figure 15.1):**

- o Unlike metals where necking leads directly to failure, semicrystalline polymers (Curve B) experience **drawing**: the polymer necks, and then the neck propagates down the entire gage length as disordered polymer chains align parallel to the tensile axis, dramatically strengthening the necked region!

4. 📚 Teacher Notes Cross-Reference & Concordia Exam Traps

- **Concordia Exam Focus:**

- o Calculating repeat unit molecular weight m and finding DP from given $\bar{M}_n$.
- o Comparing properties above and below T_g (e.g., rubber car tires shattering like glass in liquid nitrogen).
- o Distinguishing condensation (step-growth, releases water by-product) vs. addition (chain-growth) polymerization.

- **Concordia Exam Traps:**

- o **Repeat Unit Mass (m) Calculation:** Forgetting to account for double bonds opening up. Polyethylene monomer is ethylene (C_2H_4), so $m = 2(12.011) + 4(1.008) = 28.05 \text{ g/mol}$. In vinyl chloride ($\text{C}_2\text{H}_3\text{Cl}$), you replace one H with Cl (35.45 g/mol).

5. 💡 Master Exam Problem & Step-by-Step Solution Framework

Problem:

A polyvinyl chloride (PVC) pipe material has a number-average molecular weight $\bar{M}_n = 125,000 \text{ g/mol}$. (a) Calculate the repeat unit molecular weight m . (b) Determine its degree of polymerization DP .

- **Step 1: Determine Chemical Formula of PVC Repeat Unit**

PVC repeat unit is $[-\text{CH}_2 - \text{CHCl}-]_n$ (derived from $\text{C}_2\text{H}_3\text{Cl}$)

:

$$\text{Carbon: } 2 \times 12.011 = 24.022 \text{ g/mol}$$

$$\text{Hydrogen: } 3 \times 1.008 = 3.024 \text{ g/mol}$$

$$\text{Chlorine: } 1 \times 35.45 = 35.450 \text{ g/mol}$$

$$m = 24.022 + 3.024 + 35.450 = 62.496 \text{ g/mol}$$

- **Step 2: Calculate Degree of Polymerization DP**

$$DP = \frac{\bar{M}_n}{m} = \frac{125,000 \text{ g/mol}}{62.496 \text{ g/mol}} = 2,000.1 \approx 2,000 \text{ repeat units}$$

📖 Chapter 18: Electrical Properties

(Syllabus Week 11 · Final Exam Scope)

1. 🌀 30-Second Core Intuition & Plain-English Concept

Electrical conduction depends on whether electrons can easily jump into empty, available energy states to move freely through the solid.

Energy Band Theory:

- **Metals:** Conduction and valence bands overlap (or band is half-filled). Billions of electrons are free to move instantly $\implies$ high conductivity.
- **Insulators:** A massive forbidden energy band gap ($E_g > 5\text{eV}$) separates filled and empty states. Electrons cannot jump $\implies$ zero conductivity.
- **Semiconductors:** A narrow band gap ($E_g < 2\text{eV}$). Thermal energy can kick electrons across the gap $\implies$ conductivity increases exponentially with temperature!

2. ⚙️ High-Yield Mathematical Engine & Semiconductor Physics

1. Ohm's Law and Electrical Conductivity

$$V = IR, \quad \rho_{\text{el}} = \frac{RA}{l}, \quad \sigma_{\text{el}} = \frac{1}{\rho_{\text{el}}} = n|e|\mu_e + p|e|\mu_h$$

- n, p : Free electron and hole concentrations (extm^{-3}).
- μ_e, μ_h : Electron and hole mobilities ($\text{extm}^2/\text{V} \cdot \text{s}$).
- e : Elementary charge ($1.602 \times 10^{-19} \text{ C}$).

2. Intrinsic vs. Extrinsic Semiconductors

- **Intrinsic (Pure Si, Ge):** $n = p = n_i$. Conductivity rises exponentially with temperature:

$$\sigma \propto \exp\left(-\frac{E_g}{2k_B T}\right)$$

- **Extrinsic (Doped):**
 - **n -Type:** Doped with Group V elements (P, As, Sb). Donates free electrons ($n \gg p$).
 - **p -Type:** Doped with Group III elements (B, Al, Ga). Creates positive holes ($p \gg n$).

3. 🖼️ Textbook Reference Diagram & Visual Pedagogical Analysis

Figure 18.4 The various possible electron band structures in solids at 0 K. (a) The electron band structure found in metals such as copper, in which there are available electron states above and adjacent to filled states, in the same band. (b) The electron band structure of metals such as magnesium, in which there is an overlap of filled and empty outer bands. (c) The electron band structure characteristic of insulators; the filled valence band is separated from the empty conduction band by a relatively large band gap (>2 eV). (d) The electron band structure found in the semiconductors, which is the same as for insulators except that the band gap is relatively narrow (<2 eV).

The number of states within each band is equal to the total of all states contributed by the N atoms. For example, an s band consists of N states and a p band of $3N$ states. With regard to occupancy, each energy state may accommodate two electrons that must have oppositely directed spins. Furthermore, bands contain the electrons that resided in the corresponding levels of the isolated atoms; for example, a $4s$ energy band in the solid contains those isolated atoms' $4s$ electrons. Of course, there are empty bands and, possibly, bands that are only partially filled.

The electrical properties of a solid material are a consequence of its electron band structure—that is, the arrangement of the outermost electron bands and the way in which they are filled with electrons.

Four different types of band structures are possible at 0 K. In the first (Figure 18.4a), one outermost band is only partially filled with electrons. The energy corresponding to the highest filled state at 0 K is called the **Fermi energy** E_F , as indicated. This energy band structure is typified by some metals, in particular those that have a single s valence electron (e.g., copper). Each copper atom has one $4s$ electron; however, for a solid composed of N atoms, the $4s$ band is capable of accommodating $2N$ electrons. Thus, only half of the available electron positions within this $4s$ band are filled.

For the second band structure, also found in metals (Figure 18.4b), there is an overlap of an empty band and a filled band. Magnesium has this band structure. Each

Fermi energy

Figure 18.4: Electron band structures: (a, b) Metals (overlapping / partially filled), (c) Insulators (wide gap), and (d) Semiconductors (narrow gap).

In-Depth Visual Breakdown:

- **Figure 18.4:** Demonstrates why metals conduct and insulators do not:
 - In metals (a, b), the Fermi level E_F lies inside a band or between overlapping bands. Adjacent unoccupied states exist immediately above E_F ; negligible electrical field excites electrons into motion.
 - In insulators (c), the valence band is completely full and separated from the empty conduction band by a wide forbidden gap ($E_g > 5 \text{ eV}$). Room temperature thermal energy ($k_B T \approx 0.026 \text{ eV}$) is completely incapable of exciting electrons across this gap.
 - In semiconductors (d), $E_g \approx 1.1 \text{ eV}$ (Silicon). Thermal vibrations easily kick a steady stream of electrons into the conduction band, leaving equal numbers of conducting holes behind.

4. ⚖️ Teacher Notes Cross-Reference & Concordia Exam Traps

- **Concordia Exam Focus:**
 - Explaining the opposing temperature effects:
 - In **metals**, electrical conductivity **decreases** as temperature rises (thermal lattice vibrations scatter electrons: resistivity $\rho = \rho_0 + \alpha T$).
 - In **intrinsic semiconductors**, electrical conductivity **increases exponentially** with temperature (thermal energy excites vastly more charge carriers across E_g).

📖 Chapter 19: Thermal Properties

(Syllabus Week 11 · Final Exam Scope)

1. 🌀 30-Second Core Intuition & Plain-English Concept

Heat is simply atomic vibrations. In solids, heat is stored in quantized vibrational waves called **phonons** and transported through the material by both phonons and free electrons.

Why Materials Expand When Heated: Thermal expansion is a direct consequence of the **asymmetry (anharmonicity)** of the interatomic potential energy well! Because it takes less energy to push atoms apart than to squeeze them together, heating an atom makes it vibrate outward, increasing the average interatomic spacing.

2. ⚙️ High-Yield Mathematical Engine & Thermal Properties

1. Heat Capacity (C_v)

$$C_v = \left(\frac{\partial E}{\partial T} \right)_v$$

- At high temperatures ($T > \theta_D$, Debye Temperature), heat capacity reaches the universal **Dulong-Petit Limit**:

$$C_v \approx 3R \approx 25 \text{ J/mol} \cdot \text{K}$$

2. Linear Thermal Expansion

$$\frac{\Delta l}{l_0} = \alpha_l \Delta T$$

- α_l : Linear coefficient of thermal expansion (extK^{-1} or $^{\circ}\text{extC}^{-1}$).
- Ceramics have lower α_l than metals; polymers have the highest α_l (weak van der Waals bonds).

3. Thermal Conductivity (k) and Thermal Shock Resistance (TSR)

$$q = -k \frac{dT}{dx} \quad (\text{Fourier's Law})$$

- Thermal Shock Resistance (TSR):** The ability of a ceramic to withstand rapid cooling without fracture:

$$TSR \cong \frac{\sigma_f \cdot k}{E \cdot \alpha_l}$$

(To resist cracking during quenching, a ceramic must have high strength σ_f , high thermal conductivity k , low stiffness E , and low thermal expansion α_l)

3. 📖 Textbook Reference Diagram & Visual Pedagogical Analysis

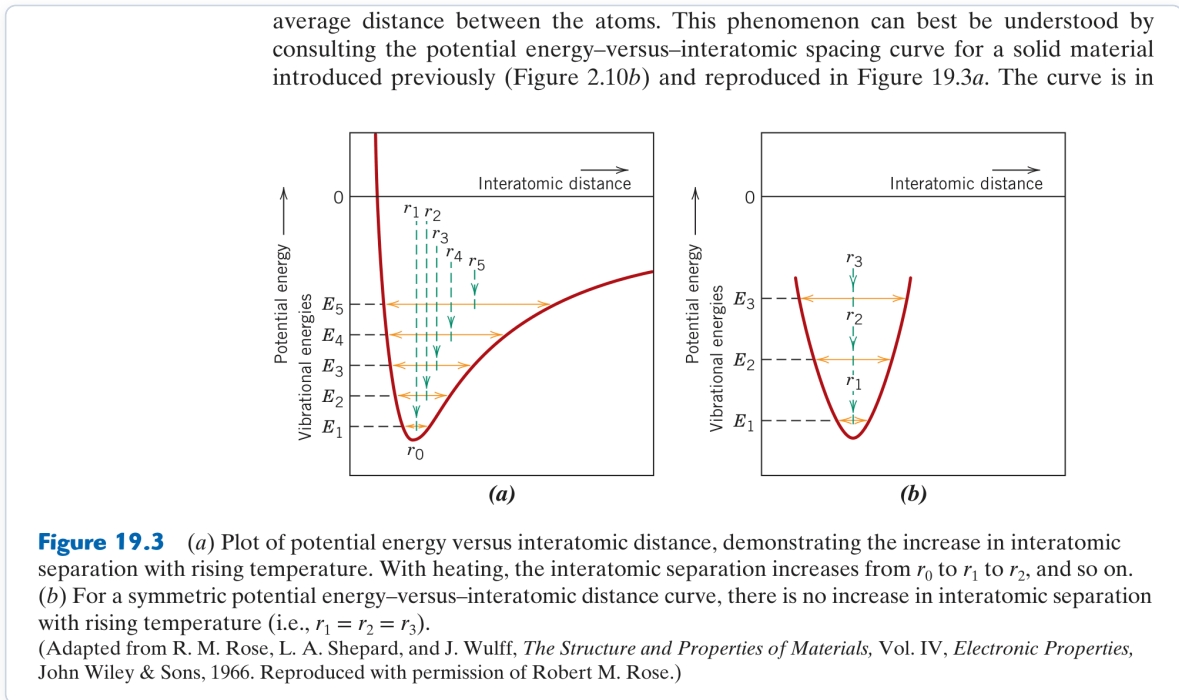


Figure 19.3: Potential energy vs. interatomic distance: (a) Asymmetric well demonstrating increase in average interatomic spacing r_T with rising temperature, vs. (b) A hypothetical symmetric well with zero thermal expansion.

In-Depth Visual Breakdown:

- **The Asymmetric Well (Figure 19.3a):** At $0\text{ext}K$, atoms sit at minimum r_0 . As temperature rises to T_1, T_2, T_3 , the atom vibrates between the left and right walls of the well. Because the repulsive wall (left) is much steeper than the attractive wall (right), the midpoint of vibration shifts progressively outward ($r_3 > r_2 > r_1 > r_0$). This macroscopic shift is thermal expansion.
- **The Hypothetical Symmetric Well (Figure 19.3b):** If interatomic forces were perfectly symmetric (parabolic well), heating would expand vibrations equally in both directions, the average separation would remain constant ($r_T = r_0$), and the material would have **zero thermal expansion** ($\alpha = 0$).

📖 Chapter 21: Optical Properties

(Syllabus Week 12 · Final Exam Scope)

1. 🎯 30-Second Core Intuition & Plain-English Concept

Light is electromagnetic radiation. When a light wave strikes a solid, three things can happen: it can bounce off (**Reflection**), get absorbed by electrons (**Absorption**), or pass through unscathed (**Transmission**):

$$R + A + T = 1$$

Why Glass is Transparent and Metals are Opaque:

- **Metals:** Have empty electronic energy states everywhere. Photons of
all

visible wavelengths are instantly absorbed by electrons and re-emitted $\implies$ metals are totally opaque and highly reflective.

- **Insulators / Glasses:** Have an energy band gap ($E_g > 3.1\text{ eV}$) that is larger than visible light photon energies ($1.8 - 3.1\text{ eV}$). Photons do not possess enough energy to excite electrons across the band gap $\implies$ light cannot be absorbed and passes straight through (transparent!).

2. ⚙️ High-Yield Mathematical Engine & Optical Physics

1. Photon Energy and Wavelength Relation

$$E = h\nu = \frac{hc}{\lambda}$$

- h : Planck's constant ($4.136 \times 10^{-15} \text{ eV} \cdot \text{s} = 6.626 \times 10^{-34} \text{ J} \cdot \text{s}$).
- c : Speed of light ($3.0 \times 10^8 \text{ m/s}$).
- **Concordia Shortcut:**

$$E(\text{eV}) = \frac{1.24}{\lambda(\mu\text{m})}$$

2. Fundamental Absorption Condition

A photon can be absorbed by an electron transition across a band gap if and only if:

$$h\nu \geq E_g \iff \lambda \leq \frac{hc}{E_g}$$

- If $E_g > 3.1 \text{ eV}$ ($\lambda < 0.4 \mu\text{m}$, ultraviolet), visible light ($0.4 - 0.7 \mu\text{m}$) cannot be absorbed, making the material **optically transparent** (e.g., Diamond $E_g = 5.5 \text{ eV}$).

3. 🖼️ Textbook Reference Diagram & Visual Pedagogical Analysis

Optical Properties of Metals

Consider the electron energy band schemes for metals as illustrated in Figures 18.4a and 18.4b; in both cases, a high-energy band is only partially filled with electrons. Metals are opaque because the incident radiation having frequencies within the visible range excites electrons into unoccupied energy states above the Fermi energy, as demonstrated in Figure 21.4a; as a consequence, the incident radiation is absorbed, in accordance with Equation 21.6. Total absorption is within a very thin outer layer, usually less than $0.1 \mu\text{m}$; thus only metallic films thinner than $0.1 \mu\text{m}$ are capable of transmitting visible light.

All frequencies of visible light are absorbed by metals because of the continuously available empty electron states, which permit electron transitions as in Figure 21.4a. In fact, metals are opaque to all electromagnetic radiation on the low end of the frequency

Figure 21.4 (a) Schematic representation of the mechanism of photon absorption for metallic materials in which an electron is excited into a higher-energy unoccupied state. The change in energy of the electron ΔE is equal to the energy of the photon. (b) Reemission of a photon of light by the direct transition of an electron from a high to a low energy state.

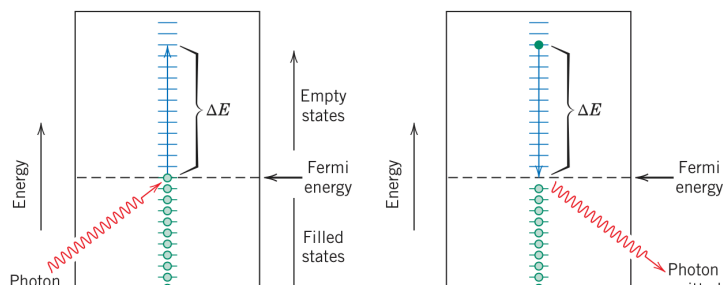


Figure 21.4: (a) Electron excitation across a band gap via photon absorption ($h\nu > E_g$), and (b) Subsequent decay and light re-emission.

In-Depth Visual Breakdown:

- **Photon Absorption (Figure 21.4a):** An incoming photon with energy $h\nu \geq E_g$ is annihilated, promoting an electron from the valence band into the conduction band and creating a hole in the valence band.
 - **Electron Relaxation (Figure 21.4b):** When the excited electron falls back to the valence band, it releases energy by emitting a photon (luminescence/fluorescence) or generating heat (phonons).
-

⚡ Master Formula & Exam Shortcut Cheatsheet

Topic	Equation	Key Units & Constants	Quick Exam Context
Bonding Energy	$E_N(r) = -\frac{A}{r} + \frac{B}{r^n}$	r in nm, E in eV	Minimum at r_0 where $dE_N/dr = 0$
Ionic Character		X : Pauling electronegativity	Ionic if
BCC Lattice	$a = \frac{4R}{\sqrt{3}}$, APF = 0.680	$n = 2$ atoms/cell, CN = 8	Atoms touch along body diagonal
FCC Lattice	$a = 2\sqrt{2}R$, APF = 0.740	$n = 4$ atoms/cell, CN = 12	Atoms touch along face diagonal
Theoretical Density	$\rho = \frac{n \cdot A}{V_c \cdot N_A}$	A in g/mol, V_c in cm ³	$N_A = 6.022 \times 10^{23} \text{ mol}^{-1}$
Miller Indices (Planes)	$(hkl) = (1/x, 1/y, 1/z) \times \text{LCD}$	Round parentheses, no commas	If plane hits origin, translate origin!
Bragg's Law (XRD)	$n\lambda = 2d_{hkl} \sin \theta$	$d_{hkl} = a/\sqrt{h^2 + k^2 + l^2}$	Remember: detector reports 2θ !
Vacancies (Arrhenius)	$N_v = N \exp(-Q_v/k_B T)$	$k_B = 8.62 \times 10^{-5} \text{ eV/K}$	Convert T to Kelvin!
Fick's First Law	$J = -D \frac{dC}{dx}$	D in m ² /s, J in kg/(m ² · s)	Steady-state diffusion flux
Carburization (Fick II)	$\frac{C_s - C_0}{C_s - C_x} = 1 - \text{erf}\left(\frac{x}{2\sqrt{Dt}}\right)$	$\text{erf}(z)$ via interpolation table	If ratio is fixed: $x^2/(Dt) = \text{const}$
Hooke's Law	$\sigma = E\epsilon$	E in GPa, σ in MPa	Elastic linear region slope
Yield Offset	0.2 strain offset	Parallel to elastic slope E	Standard σ_y determination
Schmid's Law	$\tau_R = \sigma \cos \phi \cos \lambda$	ϕ : normal to plane; λ : slip dir	Yield occurs when $\tau_R = \tau_{\text{crss}}$
Hall-Petch	$\sigma_y = \sigma_0 + k_y d^{-1/2}$	d : grain diameter in mm	Finer grains = higher σ_y and toughness
Cold Work		Cross-sectional area reduction	Increases σ_y , decreases ductility
Lever Rule	$W_L = \frac{C_\alpha - C_0}{C_\alpha - C_L}$	Opposite lever arm / Total length	Phase fraction in two-phase region
Fe - Fe ₃ C Eutectoid	$\gamma(0.76)$	Occurs at 727°C	Produces lamellar pearlite
Ceramic 3-Point Bend	$\sigma_{fs} = \frac{3F_f L}{2bd^2}$	b : width, d : depth, L : span	Brittle flexural failure stress
Polymer DP	$DP = \bar{M}_n / m$	m : repeat unit molecular weight	Chain length metric
Thermal Shock	$TSR \cong \frac{\sigma_f k}{E \alpha_l}$	High σ_f, k ; Low E, α_l	Resistance to thermal quench cracks
Photon Absorption	$E = hc/\lambda = 1.24/\lambda(\mu\text{m}) \geq E_g$	λ in μm , E in eV	Condition for electron excitation

🎯 Final Advice for MIAE 221 Exam Success

- Calculations are King:** Over 70 of the marks on Dr. Medraj's midterm and final exams come from numerical derivations (atomic packing, density, Bragg's Law, diffusion time, Schmid's Law, lever rule, and flexural strength). Practice writing out every single step with explicit SI units.

2. **Beware the Classic Traps:**

- Always convert temperature to **Kelvin** ($K = ^\circ C + 273.15$).
- In XRD, always check whether the problem gives the Bragg angle θ or the diffractometer angle 2θ .
- In $Fe - Fe_3C$ lever rule problems, always read carefully whether the question asks for **proeutectoid ferrite** or **total ferrite**.

3. **Use the Materials Science Paradigm:** When answering descriptive questions, always tie your answer back to the **Tetrahedron**: Processing $\rightarrow$ Structure $\rightarrow$ Properties $\rightarrow$ Performance. Explain the physical

why

at the atomic or dislocation level!